Managing the Rising Cost of Science: Supplier Consolidation in the Academic Life Sciences

Connie Xu (Job Market Paper), Ruby Zhang*

August 31, 2026

Abstract

Many scientific inputs are sold in increasingly concentrated markets that expose scientists to rising prices, while experimental protocols calibrated to specific products make it difficult to substitute away from them. How much science is lost when these inputs become more expensive, and which scientists are better able to adjust? Using purchase-level procurement records, we follow a merger of two large life sciences suppliers from the prices principal investigators (PIs) pay to the research they produce. Post-merger prices in affected markets rose by roughly 20 percent, while quantities declined only temporarily, indicating limited demand response. PIs facing larger increases in their input costs published less. For the average PI, input costs rose by about 2.8 percent and scientific output fell by about 1.2 percent. The consequences of higher input prices therefore extend beyond research costs to the production of science itself. The output losses fall disproportionately on early-career PIs. Financial resources do not explain this difference. Instead, the ability to adjust appears to improve with experience, consistent with PIs accumulating knowledge of how to respond to higher costs through years of running a lab.

* **Xu:** Harvard University. Email: `conniexu@g.harvard.edu`. **Zhang:** Harvard University. Email: `rzhang15@g.harvard.edu`. We are deeply grateful to Amitabh Chandra, Luca Maini, and Kyle Myers for their guidance throughout this project. We thank Leila Agha, Riley League, Chloe Tanaka, and participants at the Harvard Health Economics Seminar, the Health Policy Seminar, the Labor and Public Workshop, the American Society of Health Economists conference, and the European Association for Research in Industrial Economics conference for their many helpful comments. Erika Flynn, Bradley Scott, Jonathan Silverstone, and Octavio Diaz at GovSpend and Sean Seaver at Lab Spend generously assisted and shared their expertise with the procurement data. We gratefully acknowledge support from the Alfred P. Sloan Foundation's Early Career Fellowship on the Fiscal and Economic Effects of Innovation and Productivity Policies, awarded through the National Bureau of Economic Research (#G-2024-22690; Xu); the Lab for Economic Applications and Policy Award; the John and Kiendl Gordon Economics Support Fund; the Institute for Humane Studies (IHS020076; Zhang); the National Institute on Aging (T32-AG000186; Xu); and the National Science Foundation (DGE-2140743; Xu). The computations in this paper were run on the FASRC Cannon cluster supported by the FAS Division of Science Research Computing Group at Harvard University. The survey involved in this paper was exempted in August 2026 under Protocol #IRB26-0945.

1 Introduction

Scientific discovery is a resource-intensive process that depends not only on ideas and research talent, but also on experimental inputs purchased from outside suppliers. Many of these inputs are sold in increasingly concentrated supplier markets (Commission of the European Communities, 2008; Federal Trade Commission, 2014), exposing scientists to price increases for some of their most essential inputs. Substituting away from inputs whose price has risen is difficult because experimental protocols are calibrated to specific products, making adjustments costly to implement and scientifically risky (Bradbury and Plückthun, 2015; Freedman et al., 2015). Increases in input prices can therefore translate into higher research costs that scientists cannot easily avoid.

Even so, bearing these higher research costs does not necessarily mean producing less science. Scientists can adjust along many dimensions of the research process, including which experiments to run, whom to hire, and what to purchase and from which supplier. Scientists may differ, however, in the advantages they have in responding to higher costs. Financial resources may provide one such advantage, allowing scientists to continue their planned research. Experience may offer another advantage, as scientists accumulate over their careers tacit knowledge about experimental approaches, alternative inputs, and ways of reorganizing projects when research becomes more costly (Coppola et al., 2025). How scientists respond to higher input costs can therefore reveal not only whether these higher costs affect scientific production, but also what enables some scientists to adjust more effectively than others.

To answer these two questions, we study a rise in scientific input prices arising from supplier consolidation and measure how much scientific output is lost to the higher prices and which principal investigators (PIs) are able to adjust to them. We assemble novel purchase-level procurement records from 165 U.S. public universities spanning 2010 to 2019 and leverage the 2014 merger of Thermo Fisher Scientific and Life Technologies Corporation, two of the largest life sciences input suppliers. The merger increased concentration in markets where the two firms overlapped in supplying consumables—routinely purchased experimental materials such as reagents, enzymes, and cell culture media—generating plausibly exogenous variation in the prices PIs paid for affected consumables.

We first establish that the merger raised prices in the affected markets, and that the increase translated almost entirely into higher spending. Using a matched difference-in-differences design, we compare product markets with pre-merger competitive overlap against similar, unaffected comparison markets. Prices in affected markets rose by approximately 20 percent relative to comparison markets, while quantities showed no differential decline, with both affected and comparison markets experiencing only a temporary dip. Spending therefore rose nearly one-for-one with prices, indicating that demand for these consumables is highly inelastic. Rather than avoiding the price increase by purchasing less or substituting away, PIs continued purchasing the same consumables at higher prices, leaving research budgets to bear the higher cost.

The merger raises consumables costs differently across PIs because different experimental methods require different bundles of consumables. Our exposure measure captures this variation by weighting each market's estimated price effect by a PI's pre-merger spending share in that market. For PIs linked to procurement records, we construct exposure directly from their purchases. For the remaining PIs, we impute exposure based on the similarity of their publications to those of PIs whose purchases we observe. This approach yields a panel of 13,515 U.S. life sciences PIs, allowing us to compare the research output of more- and less-exposed PIs before and after the merger.

Our analysis finds that these higher consumables costs translate into less scientific output. Event-study estimates show a sustained decline in publications among more-exposed relative to less-exposed PIs following the merger. Affected markets account for only part of a PI's consumables bundle and face different price increases, so the average PI's consumables costs rise by approximately 2.8 percent, while publication output falls by about 1.2 percent. These magnitudes imply an elasticity of scientific output with respect to consumables costs of approximately −0.42. In dollar terms, one fewer publication corresponds to approximately \$10,500 in higher consumables costs. The decline in publications therefore suggests that PIs cannot fully adjust to the increase, allowing part of the shock originating in input markets to reach scientific output.

This output loss falls disproportionately on early-career PIs. PIs with fewer than 14 years of experience see output decline as their consumables costs rise, while the response among late-career PIs is smaller and cannot be statistically distinguished from zero. We measure experience as years since a PI's first last-authored publication, our proxy for when a PI begins running their own lab. Experience measured instead from a PI's first publication ever, which adds the years spent at the bench during training, produces a larger and statistically significant difference between early- and late-career PIs.

One potential explanation is that late-career PIs have greater financial resources with which to absorb the cost shock. Yet when we split PIs by proxies for available resources, including baseline funding from the National Institutes of Health (NIH) and the R&D environment of their institutions, we find no comparable gradient. Comparing PIs with similar baseline funding leaves the difference by career stage intact.

We interpret this pattern as consistent with late-career PIs drawing on tacit knowledge accumulated over their careers—knowledge about how to reorganize their research and purchase more efficiently when input costs rise. Preliminary evidence from our survey of currently active life sciences PIs supports the premise that this knowledge is learned by doing rather than transferred across PIs. A large majority of the 19 respondents report that they learned how to manage supply purchasing through trial and error rather than through formal training. We nevertheless view this mechanism as suggestive rather than definitive.

Taken together, the estimated price and output responses illustrate the broader incidence of supplier market power in academic science. Across our sample, a single merger implies approximately \$3.7 million per year in additional spending on consumables alongside roughly

349 fewer papers per year. The research budgets bearing this cost are funded largely by federal grants from agencies such as the NIH and the National Science Foundation (NSF). In effect, public research funds are redirected toward private suppliers to purchase largely the same consumables at higher prices, while less science is produced.

Our first contribution is to show that the production of science depends not only on scientists' access to inputs, but also on the input markets in which they buy them. Existing work asks how the amount and direction of research change when scientists gain or lose funding (Jacob and Lefgren, 2011; Li et al., 2017; Azoulay et al., 2021; Howell, 2017; Myers, 2020) or access to research materials, infrastructure, equipment, and technologies (Murray et al., 2016; Furman and Stern, 2011; Waldinger, 2016; Baruffaldi and Gaessler, 2021; McKeon and Boudou, 2024; Furman and Teodoridis, 2020). We instead examine what happens to scientific research when the prices of consumables scientists regularly use rise as input markets consolidate. Because experimental protocols tie PIs to specific products, these price increases raise the cost of conducting research and reduce scientific output. These effects can be widespread because such inputs are used throughout the experimental life sciences and supplied by a small number of firms.

Our second contribution is to show that experience shapes not only how much research a scientist produces, but also how well a scientist adjusts when research becomes more costly. A large literature documents productivity gains from experience and learning by doing, including gains that continue over the course of a career and arise through repeated experimentation (Papay and Kraft, 2015; Hendel and Spiegel, 2014; Reagans et al., 2005; Dong et al., 2024; Caplin et al., 2026), while related work shows that management practices can generate large productivity differences across firms (Bloom and Van Reenen, 2007; Bloom et al., 2013). We show that these forces also matter for adjustment to cost shocks. Output declines are concentrated among early-career PIs, and differences in financial resources do not explain why they experience larger declines. Our findings suggest that the ability to navigate cost shocks develops in part through experience running a research lab. Early-career PIs may therefore encounter input-market disruptions before developing this ability.

Our third contribution is to show that the costs of supplier market power can propagate beyond the market where it is exercised. Merger retrospectives typically measure the effects of consolidation through prices paid by buyers (Ashenfelter et al., 2014; Dafny et al., 2012), while work on public spending examines whether market power diverts public dollars from intended beneficiaries to private firms (Duggan et al., 2016; Cabral et al., 2018). We trace the effects further downstream to the research those purchases support and show that PIs more exposed to higher input prices publish fewer papers. Unlike firms producing market goods, PIs cannot generally pass higher input costs on through the price of scientific knowledge. The incidence of market power can therefore take the form of less knowledge produced rather than higher downstream prices. This propagation is particularly consequential because knowledge spillovers may already leave research underprovided relative to its social value (Nelson, 1959; Arrow, 1962). Market power in scientific input markets can thus compound this distortion by reducing the knowledge that public research funding is intended to produce.

2 Institutional Setting

A majority of a PI's research budget is committed in advance, leaving consumable supplies as one of the few areas of expenditure that can be adjusted within a funding cycle. PIs, however, have little control over the prices they pay for the products they need, since consumables are sold in a highly concentrated industry where prices are privately negotiated and experimental protocols often depend on specific products with few close substitutes. A PI facing a price increase for a required input therefore has limited ability to avoid the increase through substitution and little flexible room in their budget to absorb it. We study how such price increases affect research spending and scientific output using Thermo Fisher Scientific's 2014 acquisition of Life Technologies.

2.1 University Scientists' Research Budgets

The total amount a PI can spend within a funding cycle cannot easily be increased. Federal grants from agencies such as the NIH and NSF fund the vast majority of academic life sciences research and are the main source PIs rely on for personnel, equipment, and supplies (Azoulay et al., 2019). These grants provide PIs with a set budget to produce research over a period that typically lasts three to five years (Stephan, 2012; Azoulay et al., 2011; Carnehl et al., 2025; Tham, 2023). Grant amounts are generally not renegotiated within the funding cycle, so PIs have limited ability to increase their budgets when costs rise. A PI can seek additional funding, but new grants require time to secure and can arrive only in later funding cycles (Jacob and Lefgren, 2011).

Within a grant cycle, most of the direct costs are committed to areas a PI cannot easily adjust. Personnel costs (e.g., PI summer salaries, postdocs, graduate students, technicians) represent roughly 80 percent of a typical NIH research budget and cannot be cut on short notice without breaking multi-year commitments (Stephan, 2012; National Institute of Allergy and Infectious Diseases, 2026). Supplies and equipment together account for the remaining 20 percent. Within this category, consumables (e.g., reagents, plasticware, kits) make up the majority, roughly 10 to 20 percent of the budget, because large instrumentation such as freezers and microscopes is a front-loaded cost, often funded institutionally, and shared across research groups (Ehrenberg et al., 2003).

Consumables are therefore the margin a PI can most readily adjust within an active grant cycle, yet that flexibility is also constrained. Substitutes are limited at the lab bench, so a higher price is difficult to avoid by switching products. A PI facing higher consumables prices is therefore likely to give something up—running fewer experiments, deferring hiring to later cycles, or diverting time from the bench to raising additional funds. Each of these responses affects the production of research itself, so variation in the price of consumables is a channel through which supplier behavior can reach scientific output.

2.2 Market Structure of the Life Sciences Supplier Industry

The life sciences supplier market concentrates pricing power in a handful of firms. Three structural features of this industry keep prices high and leave buyers with little recourse: high concentration met with little buyer power, opaque pricing, and high switching costs.

Years of consolidation have left the life sciences supplier market dominated by a few full-line firms, namely Thermo Fisher Scientific, Avantor, and MilliporeSigma, that both manufacture their own products and operate the distribution networks reselling other manufacturers' supplies to university PIs (Berkrot, 2014). Smaller independent manufacturers, such as New England Biolabs, make their own reagents but typically reach a broad customer base by selling through the distribution networks of these full-line firms. Thermo Fisher Scientific alone reported about $44.6 billion in revenue in 2025, over 42 percent of it from consumables and laboratory products (Thermo Fisher Scientific Inc., 2026), and Avantor and MilliporeSigma each report several billion dollars in annual sales of laboratory chemicals, reagents, and other consumables (Avantor, Inc., 2026; Merck KGaA, 2025). These firms combine in-house manufacturing with broad distribution catalogs, making a small number of vendors the primary channels through which PIs purchase consumables and equipment. In our conversations with procurement offices at several universities, most reported relying heavily on Thermo Fisher Scientific and Avantor, whose catalogs are often integrated directly into university purchasing platforms. This concentration of purchasing gives a few vendors substantial pricing power, with limited countervailing power on the buyer side. Universities typically negotiate volume discounts, but contracting is often decentralized across schools or departments, and even consortia that pool demand across institutions negotiate discounts from list prices that suppliers themselves set and can raise. Academic buyers also account for a relatively small share of suppliers' revenues compared with biopharmaceutical firms, further limiting their leverage. Hospitals and pharmacies, by contrast, can secure lower prices for devices and drugs by steering volume across competing suppliers (Grennan, 2013; Ellison and Snyder, 2010). Universities have less scope to do so because PIs' experimental protocols often determine which products they purchase, which restricts their ability to shift demand across suppliers.

Opaque pricing keeps buyers from knowing whether the prices they pay are competitive. Each university negotiates a private contract that sets discounts off list price, so two otherwise similar universities can unknowingly pay very different prices for the same reagent (Innovative Bioscience, n.d.). This opacity extends within institutions, as individual PIs can negotiate separately with sales representatives, so prices differ even across PIs in the same department (Donna Kridelbaugh, 2018). Opaque pricing of this kind has been studied in other industries, where comparable buyers pay widely dispersed prices for the same product (Grennan, 2013), and disclosing to buyers what their peers pay lowers the prices those buyers subsequently pay (Grennan and Swanson, 2020; Brown, 2019). This evidence indicates that opacity itself sustains higher prices by preventing buyers from benchmarking, and that suppliers have incentives to keep price information hidden.

Even when cheaper alternatives exist, PIs face high switching costs at the bench. Experimental protocols are calibrated to specific reagents, instruments, and kits (Walsh et al., 2007, 2005), so adopting a new supplier's product requires revalidation that can take weeks or months and risks results that do not reproduce (Torrington, 2021; Lorsch et al., 2014; Rehman, 2022). Reagent variability is itself a well-documented source of irreproducibility (Bradbury and Plückthun, 2015; Weber et al., 2022), and surveys attribute a substantial share of failed replications to the sensitivity of biological reagents (Baker, 2016; Freedman et al., 2015). Comparing prices across suppliers is also costly, since doing so takes time away from research. In addition, once a PI has sunk the cost of building a protocol around a specific reagent, switching requires incurring new costs to adopt and validate an alternative, making the PI more likely to continue purchasing the same reagent as its price rises. Demand for consumables is therefore likely to be highly inelastic, and PIs remain locked into their incumbent suppliers.

Taken together, high concentration, opaque pricing, and high switching costs create an environment where a small number of suppliers face buyers whose demand is highly inelastic. A firm that gains market power in this setting can therefore raise prices with little risk of losing customers.

2.3 The Thermo Fisher Scientific–Life Technologies Merger

We study one large shock to the supplier market, Thermo Fisher Scientific's acquisition of Life Technologies Corporation, which combined two of the largest suppliers of life sciences inputs into a single firm.

Thermo Fisher Scientific was, at the time of the merger, the largest supplier of scientific instruments and consumables, with a distribution network reaching most U.S. university PIs. Life Technologies held leading positions in molecular biology and genomics reagents, including enzymes, kits, and cell-culture products. Both ranked among the few full-line suppliers able to cover a wide range of life sciences laboratory inputs, so their combination raised concentration across the consumables markets they jointly supplied. Thermo Fisher Scientific agreed to acquire Life Technologies for approximately $13.6 billion in April 2013 and completed the transaction on February 3, 2014 (Federal Trade Commission, 2014).

Competition authorities worldwide reviewed the transaction and scrutinized the extent of the merging firms' competitive overlap. The Federal Trade Commission estimated that the combined firm would hold at least 50 percent of the worldwide market for cell culture media, about 60 percent for cell culture sera, and over 50 percent of gene-modulation (siRNA) reagents (Federal Trade Commission, 2014). The authorities cleared the deal only after Thermo Fisher Scientific divested the two overlapping lines, HyClone cell culture and Dharmacon gene-modulation, to GE Healthcare for approximately $1.06 billion.[1] The divestiture, however, transferred only the manufacturing rights to these lines, and Thermo Fisher Scientific remained free to sell the

[1] The European Commission additionally required divestiture of a polymer-based magnetic-beads business, which was included in the same sale to GE Healthcare. See the European Commission's clearance decision in Case COMP/M.6944 and the Federal Trade Commission's analysis of the consent order (File No. 131-0134).

products through its distribution network.

3 The Effect of the Merger on Prices and Spending

Our analysis shows that the merger raised prices in the treated markets by approximately 20 percent, while quantities did not fall relative to control markets, so spending in the treated markets rose with prices. Demand for these consumables is therefore highly inelastic, and research budgets bear the higher cost.

Our empirical analysis proceeds in two parts. This section estimates how the merger changed the prices of laboratory inputs. We build a panel of procurement data at the product-market, university, and year level, where a product market is a set of inputs interchangeable at the lab bench, and estimate a separate price effect for each affected market. Section 4 then traces these market-level price effects to research output, accounting for each PI's pre-merger consumables bundle so that PIs concentrated in the hardest-hit markets face the largest cost increase.

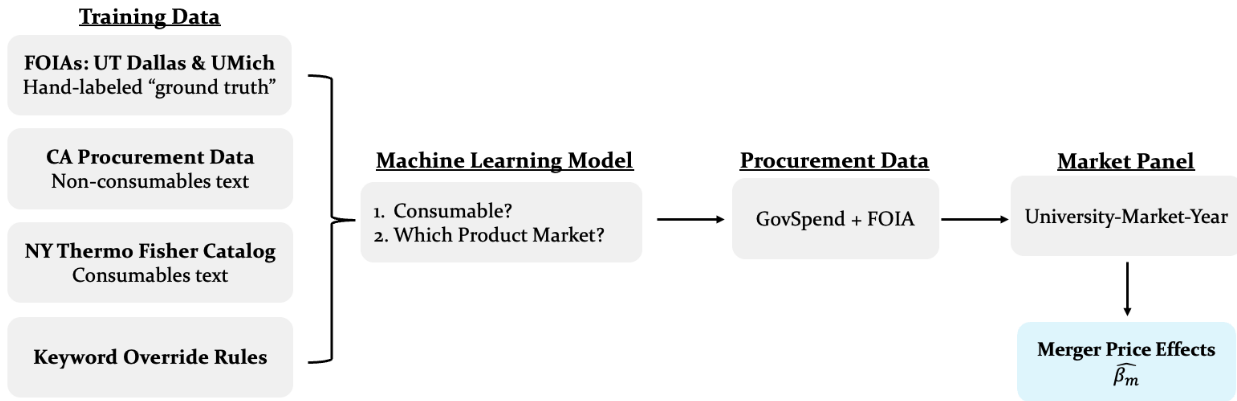
3.1 Data

Our procurement data record individual purchases at U.S. public universities and the prices paid for them. We classify each purchase into a product market and aggregate the classified purchases into the university-market-year panel of prices used to estimate the price effects. Figure 1 provides a high-level overview of this construction, in which a natural language processing (NLP) classifier assigns each item to a market from its text description.

3.1.1 Procurement Purchases

Our procurement data come from a commercial data vendor covering purchase records at 165 public universities and from Freedom of Information Act (FOIA) records we filed directly with public universities. The FOIA records supplement the price data and provide training data for

Figure 1. Construction of the Market Panel



Notes: The two-step classifier is trained on hand-labeled FOIA procurement records from UT Dallas and UMich, supplemented with California procurement records, a Thermo Fisher Scientific catalog, and keyword override rules. Applying it to the GovSpend extract sorts consumables purchases into 1,105 product markets and produces the university-market-year procurement panel used in this section.

our NLP classifier that assigns line-item purchases to their relevant product market. Since public records laws govern which purchase records are disclosable, the transactions we observe largely reflect federally funded purchases, though we cannot verify the funding source of individual transactions, and some may be privately funded.

GovSpend. The commercial data vendor, GovSpend, provides us with 4.8 million life sciences purchases across 165 public universities between 2010 and 2019, where each line-item record reports the item description, price, quantity, total spending, purchase date, supplier name, and purchasing university. GovSpend is a procurement data platform that aggregates publicly available purchase records from U.S. public entities through open records requests and partnerships with institutional purchasing portals. We build our extract by focusing on public universities and applying over 1,500 keyword filters designed to capture life sciences items. These data, however, do not report the unit size of the item purchased, such as a 50 mL versus a 100 mL bottle of fetal bovine serum. Our price measure is therefore the average price per purchase rather than the price per unit, and our quantity measure is the number of purchases rather than the number of units. We return to this limitation in Section 3.3.

FOIA Requests. FOIA records serve two purposes in our price analysis. First, records from Oregon State University provide item-level price data that supplement the GovSpend data.

Second, records from the University of Texas at Dallas (UT Dallas) and the University of Michigan (UMich) form the basis of our training data. UT Dallas records include item-level catalog numbers, which allow us to precisely identify products, while the UMich records offer broad coverage across product types. We hand-label each item in these two datasets to construct training data for our classifier.

3.1.2 Classification of Items into Product Markets

Our measure of prices is within a *product market*, which we define as a set of inputs that are interchangeable at the lab bench. We classify purchases into these markets by building an NLP classifier trained on hand-labeled data and applying it to our procurement data.

Labeled Training Data. Our training labels come from manually classifying the procurement records we obtained through FOIA requests from the life sciences departments at UT Dallas (2011–2024, N=61,882) and UMich (2010–2019, N=306,883).[2] As the largest full-line supplier, Thermo Fisher Scientific offers a product catalog that spans nearly the entire range of life sciences inputs, so we use its catalog hierarchy as the backbone taxonomy for our product markets and map items from smaller suppliers onto the closest Thermo Fisher Scientific category. For each unique item description–supplier pairing, we look the item up in its supplier's product catalog to identify what it is, then assign it to the narrowest Thermo Fisher Scientific category that applies. For example, this methodology separates *Taq DNA Polymerase*, *Hot Start DNA Polymerase*, and *High-Fidelity Hot Start DNA Polymerase* into distinct markets rather than a single *DNA Polymerase* category, matching their distinct use cases in polymerase chain reaction (PCR) protocols. In certain cases, the classifier cannot resolve a broad category into subcategories that are interchangeable at the lab bench from the item description alone, so we retain a coarser market rather than impose splits the descriptions cannot support. *Primary Antibodies*, for example, target different antigens and are not substitutes at the bench, yet their descriptions

[2] UT Dallas covers the Departments of Biological Sciences and Chemistry & Biochemistry, while UMich covers the Departments of Molecular, Cellular and Developmental Biology, Biological Chemistry, and Pharmacology.

are too similar to separate reliably. This coarsening is concentrated in markets that are neither treated nor selected as matched control markets, so it has little scope to influence our estimated merger effects. After assigning item descriptions to product markets, we also code each product market as being either a consumable or a non-consumable. This labeling effort yields 368,765 item descriptions mapped to 2,108 product markets.

Classification Algorithm. Our NLP classifier operates in two steps. The first step separates consumables from non-consumables by combining deterministic keyword rules with a logistic regression on term frequency-inverse document frequency (TF-IDF) text representations of the item descriptions.[3] The keyword rules classify obvious cases directly, treating descriptions that contain *polymerase* or *antibody* as consumables and those that contain *publishing* or *furniture* as non-consumables, while the logistic regression classifies the remainder. The second step assigns each consumable to one of the 1,105 consumables product markets in our labeled training data using centroid-based nearest-neighbor matching in TF-IDF space, refined with keyword override rules for item descriptions that contain unambiguous market identifiers. These override rules are particularly useful when two product markets share many common keywords. For example, items containing "taq" and "polymerase" but not "hot start" or "high-fidelity" are assigned to *Taq DNA Polymerase* rather than *Hot Start DNA Polymerase* or *High-Fidelity Hot Start DNA Polymerase*, regardless of cosine similarity score.

We train both steps on an 80/20 train/validation split of the labeled UT Dallas and UMich data, supplemented for non-consumables coverage with California state procurement records (2012–2015)[4] and Thermo Fisher Scientific's negotiated price list with New York State filtered to non-consumables purchases. Table 1 reports the classifier's precision and recall on the held-out UT Dallas and UMich records. The first step separates consumables from non-consumables

[3] We ran a BERT-based alternative and found that the TF-IDF text representations outperformed at substantially lower computational cost.
[4] These records come from the California Open Data Portal. Each purchase order carries a United Nations Standard Products and Services Code (UNSPSC), a standardized taxonomy that classifies products and services, and we retain only orders whose UNSPSC codes fall in non-consumable categories.

reliably. The second step sorts consumables into many fine-grained markets, so its precision and recall, averaged equally across markets, are lower, especially where labeled examples are scarce. Support therefore matters for accuracy, and we restrict the estimation panel to well-supported markets in Section 3.1.3.

Table 1. Classifier Performance on Held-Out Purchase Orders

	Precision	Recall	N
Step 1: Consumables Versus Non-consumables Classifier			
Consumables	0.96	0.98	31,551
Non-consumables	0.93	0.85	8,324
Step 2: Product Market Classifier			
All consumables categories	0.54	0.70	295,622
Large Markets (≥ 25 training examples)	0.74	0.73	291,394

Notes: Ground-truth labels come from the UT Dallas and UMich expenditure files. Reported rows are held out from training, and N gives per-step counts. Step 2 conditions on true consumables and reports how often the correct product market is recovered, with precision and recall averaged equally across markets so that low-support markets weigh as heavily as high-support ones. Large Markets restricts to categories with at least 25 training examples, covering 98.6 percent of true consumables.

3.1.3 Final Estimation Panel

We impose two restrictions on the procurement panel to ensure precise estimation of market-level price effects. To keep the panel's composition stable, we keep only the 76 universities that report consistently from 2010 to 2019,[5] which leaves 1.2 million consumables purchases after we apply our classifier to the GovSpend extract, each assigned to one of the 1,105 consumables product markets in our labeled training data. To minimize measurement noise, we retain only the product markets the classifier assigns with high confidence, defined as at least 25 training examples and precision and recall of at least 0.8, because both fall sharply when training examples are scarce. The resulting high-confidence markets account for 56 percent of total consumables spending ($114 million), and Appendix Table A1 reports the universities, suppliers, and product markets with the largest shares of that spending.

[5] A university is defined as reporting consistently if its purchase records cover at least eight months of each year.

3.2 Empirical Strategy

We estimate a separate merger-induced price effect for each treated market with a difference-in-differences design that compares it to matched control markets.

Defining treated and control markets. A product market is treated ($D_m^T = 1$) if it appears in the European Commission's clearance decision (Case COMP/M.6944) as a market in which Thermo Fisher Scientific and Life Technologies Corporation had pre-merger competitive overlap, and markets not discussed in the decision are classified as control ($D_m^T = 0$). We rely on the Commission's market definitions, which draw on customer-substitution evidence, reconstructed pre-merger market shares, transaction data, and the parties' internal documents, rather than building concentration measures from our own procurement data.[6]

The treated markets span the core molecular biology workflow, including cell culture media and sera (e.g., DMEM, fetal bovine serum) and nucleic-acid amplification enzymes and ready-to-use kits (PCR, qPCR, RT-PCR). Control markets cover consumables outside the merging firms' direct overlap, such as glassware, bulk lab chemicals, and pipette tips. Of the 1,105 consumables product markets in our labeled training data, 231 are treated under this definition. Restricting to the high-confidence estimation subset leaves 46 treated and 106 control markets. Appendix B lists all high-confidence markets by treatment status.

Empirical Specification. We use a difference-in-differences specification to estimate the merger's effect on three logged market-level outcomes: (1) average purchase price, (2) total quantity purchased, and (3) total spending. Together they identify both the size of the price shock and the purchasing response to it. For each treated market m^*, we construct a set of matched controls of up to two markets using nearest-neighbor matching on pre-period log

[6] A natural alternative would be to compute the Herfindahl-Hirschman Index (HHI) directly from purchasing records, since HHI captures pre-merger market concentration. Implementing this in our setting is complicated by the fact that GovSpend records supplier names as free-text fields rather than standardized identifiers, so the same firm often appears under multiple variants ("Thermo Fisher Scientific," "Thermo Fisher Sci.," and "ThermoFisher"). While manual cleaning of these fields helps, residual noise still propagates into any firm-level market share calculation and makes HHI a noisy proxy for market concentration.

price trends and averages, sampling with replacement so that the same market can serve as a matched control for multiple treated markets.[7] This matching process yields 51 distinct control markets across the 46 treated-market comparisons. Our difference-in-differences specification is as follows:

$$y_{jmt} = \beta_m \cdot D_m^T \cdot \text{Post}_t + \gamma_j + \delta_m + \lambda_t + \epsilon_{jmt}, \tag{1}$$

where y_{jmt} is the log outcome (price, quantity, or spending) in product market m purchased by university j in year t. The terms γ_j, δ_m, and λ_t are university, market, and year fixed effects. The indicator D_m^T equals one for treated markets, and $\text{Post}_t = \mathbf{1}\{t \geq 2014\}$ indicates the post-merger period. Observations are weighted by 2013 market spending, and standard errors are clustered at the market level. We estimate equation (1) separately for each of the 46 treated markets, producing market-level price effects $\{\hat{\beta}_m^P\}_{m=1}^{46}$ and analogously $\{\hat{\beta}_m^Q\}$ and $\{\hat{\beta}_m^{PQ}\}$ for quantity and spending. These market-level effects are noisily estimated, so we apply normal–normal empirical Bayes shrinkage to the $\hat{\beta}_m^P$, shrinking each estimate toward a precision-weighted cross-market mean in proportion to its sampling variance, and use the shrunk estimates throughout the paper. Shrinkage is mild, with a correlation of 0.996 between raw and shrunk estimates and an average move of 0.011 log points.

Identification rests on the common trends assumption that, absent the merger, treated and control markets would have followed the same counterfactual trajectory in each outcome. We assess the validity of this assumption by examining pre-trends in an equivalent event study specification shown in Section 3.3.

[7] Before matching, we remove from the pool of candidate control markets any market subject to its own contemporaneous shock over the sample period, so that no control carries a price movement unrelated to the merger. Examples include laboratory filtration (Danaher's 2015 acquisition of Pall) and synthetic DNA oligonucleotides (the entry of Twist Bioscience and the scaling of Integrated DNA Technologies). These 12 markets remain in the descriptive sample but are excluded from the candidate controls used to estimate the price effects.

3.3 Price, Quantity, and Spending Results

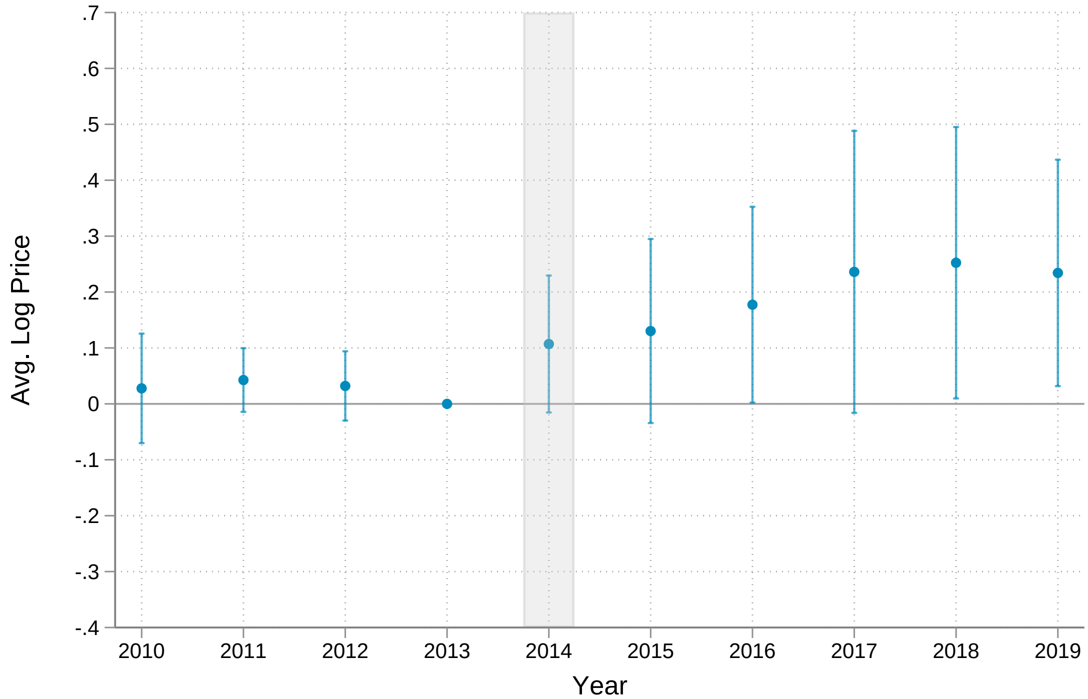
Treated product markets experienced persistent and growing price increases following the merger while their quantities did not fall relative to controls, so spending in the treated markets rose with prices. Figure 2 presents the price series in two panels: Panel (a) plots the pooled difference between treated and control markets, and Panel (b) plots the treated and control price trajectories separately, each normalized to the year before the merger. The pooled difference in Panel (a) jumps in 2014 and climbs steadily, from roughly 11 percent in the first post-merger year to 23 percent by 2019, averaging roughly 20 percent over the post-merger period, and its flat pre-merger coefficients are consistent with the common-trends assumption behind equation (1). Panel (b) traces this difference to the treated markets, whose prices climb steadily above their pre-merger baseline while control prices show no movement.

We find no evidence of anticipatory responses. Appendix Table A2 reports a placebo test that assigns a false 2012 treatment date to the pre-merger panel and shows no placebo effects on prices or quantities. While the spending estimate is marginally significant, its negative sign runs counter to the post-merger increase.

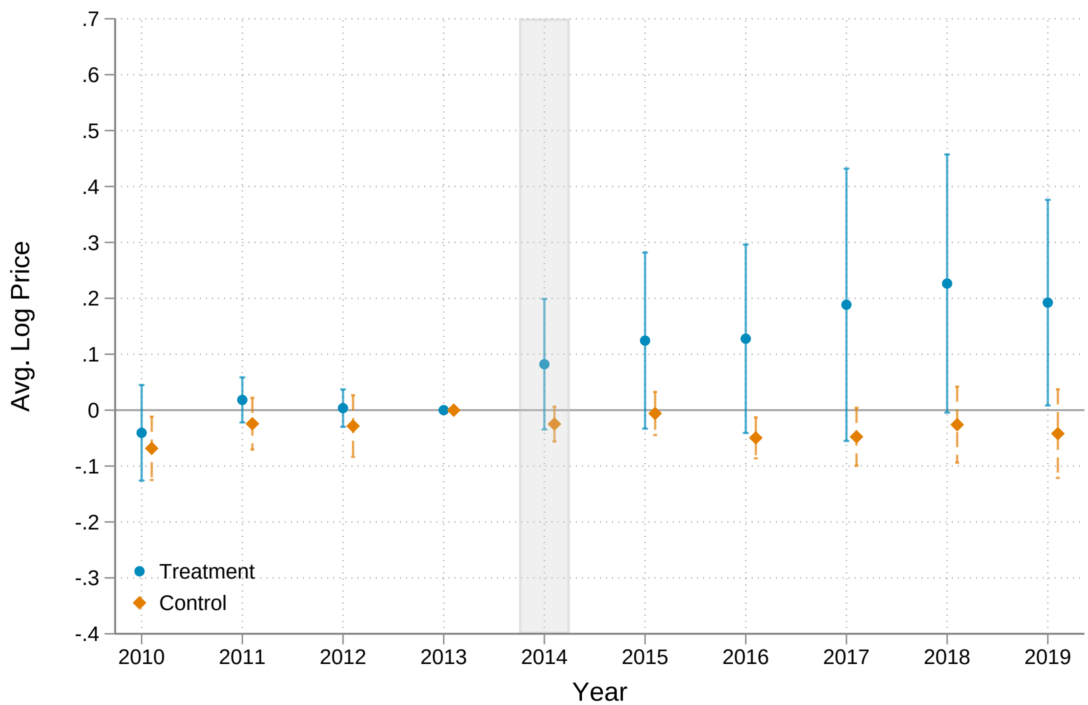
Treated quantities do not fall relative to controls after the merger, suggesting that demand for the treated inputs is highly inelastic. Figure 3 presents the quantity response in two panels, plotting the pooled difference between treated and control quantities in Panel (a) and the two series separately in Panel (b). The pooled difference in Panel (a), which isolates whether treated quantities fell relative to controls, remains statistically indistinguishable from zero in every year. The separate series in Panel (b) show a dip in both treated and control markets, each falling in the first year after the merger before returning to its pre-merger level by the end of the sample. This movement is common to both treated and control markets and therefore nets out of the pooled difference. One interpretation is a temporary tightening of research budgets following the merger, which reduced purchases across all consumables markets before they recovered. What matters for substitution, however, is whether treated quantities fell relative to controls as

Figure 2. Pooled Merger-Induced Price Effects

(a) Pooled difference



(b) Treated and control series



Notes: Event study estimates of average log prices around the Thermo Fisher Scientific–Life Technologies merger. Panel (a) plots the pooled difference between treated product markets and their matched controls. Panel (b) plots treated product markets (blue) and their matched controls (orange) separately. The vertical gray band indicates the merger year (2014). Coefficients are normalized to zero at $t = -1$. The average price in $t = -1$ is $143.30 in treated markets and $104.07 in matched control markets.

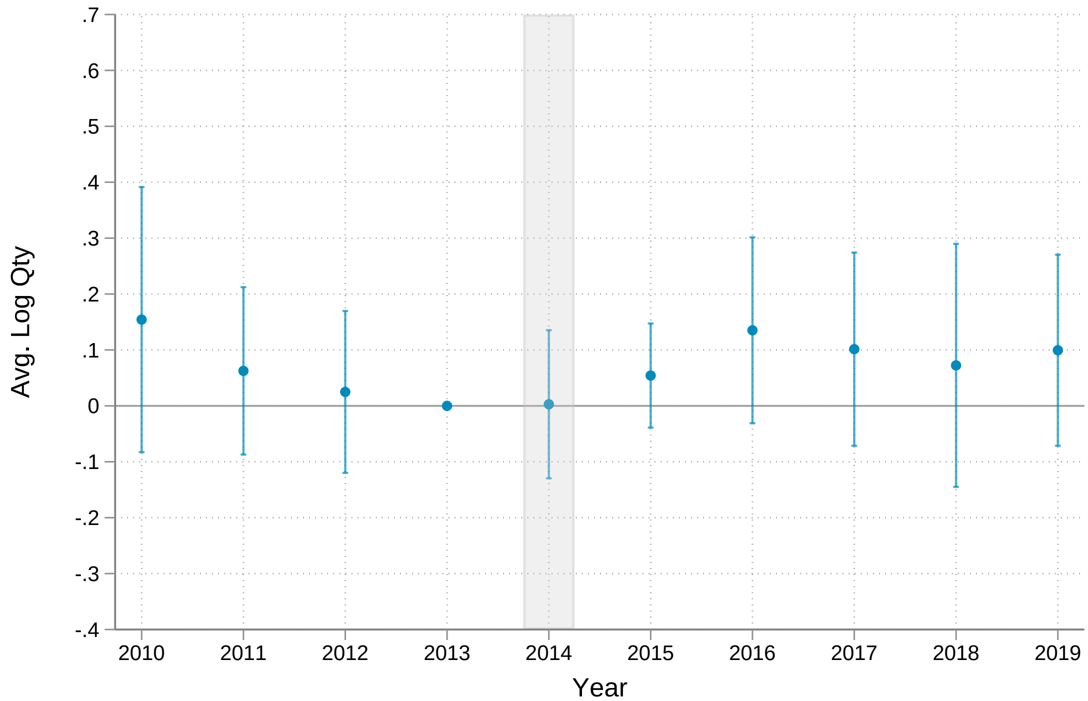
their relative prices rose. They did not, consistent with the difficulty of substituting away from specific products.

Spending decomposes as price times quantity, so with no differential quantity response the pooled spending difference in Panel (a) of Figure 4 roughly matches the 20 percent price effect in Figure 2. Panel (b) shows the same pattern in the separate series, as treated spending climbs steadily after the merger while control spending shows no sustained rise.

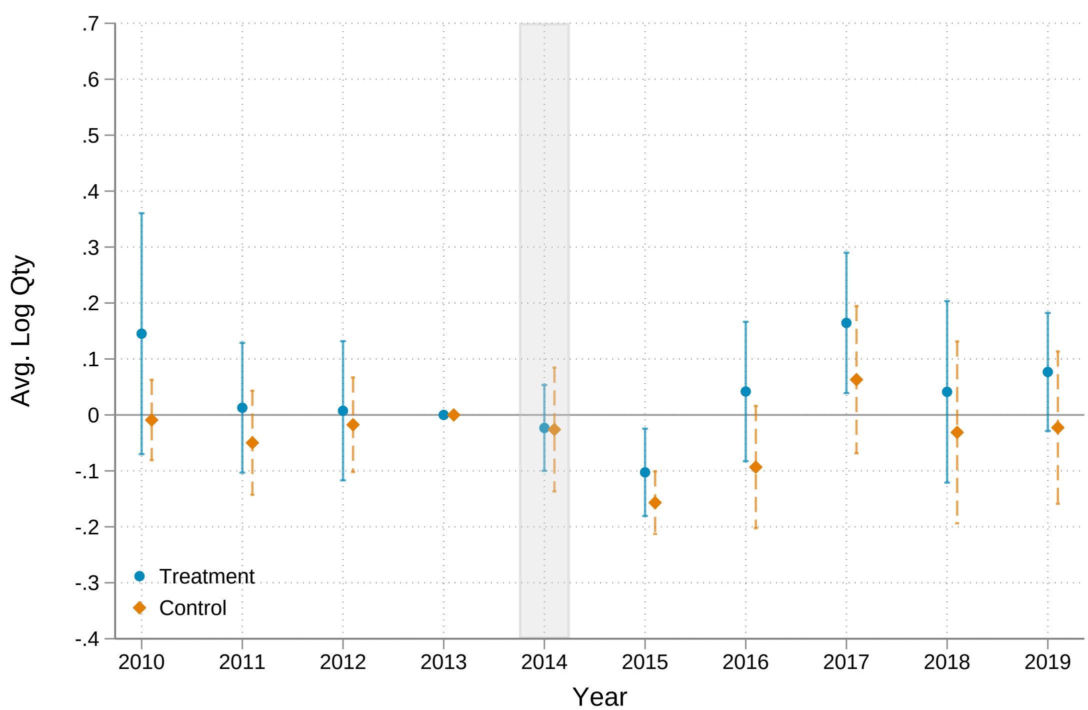
Together, these three series characterize demand for the treated consumables as highly inelastic. Treated quantities barely move relative to controls despite the roughly 20 percent price increase, so the higher price passes almost entirely into higher spending on the same essential inputs rather than into substitution toward cheaper alternatives or lower volumes. Table 2 reports the pooled difference-in-differences coefficient for each outcome, and the estimates are stable across alternative fixed-effects specifications. Randomization inference on the market-level price effects, presented with Figure 5 below, rejects the absence of a merger price effect ($p < 0.01$). Appendix Figures A1, A2, and A3 reproduce the three event studies without matching, comparing treated markets to the full set of controls, and the less comparable pre-merger trajectories there motivate our matched-control design.

Figure 3. Pooled Merger-Induced Quantity Effects

(a) Pooled difference



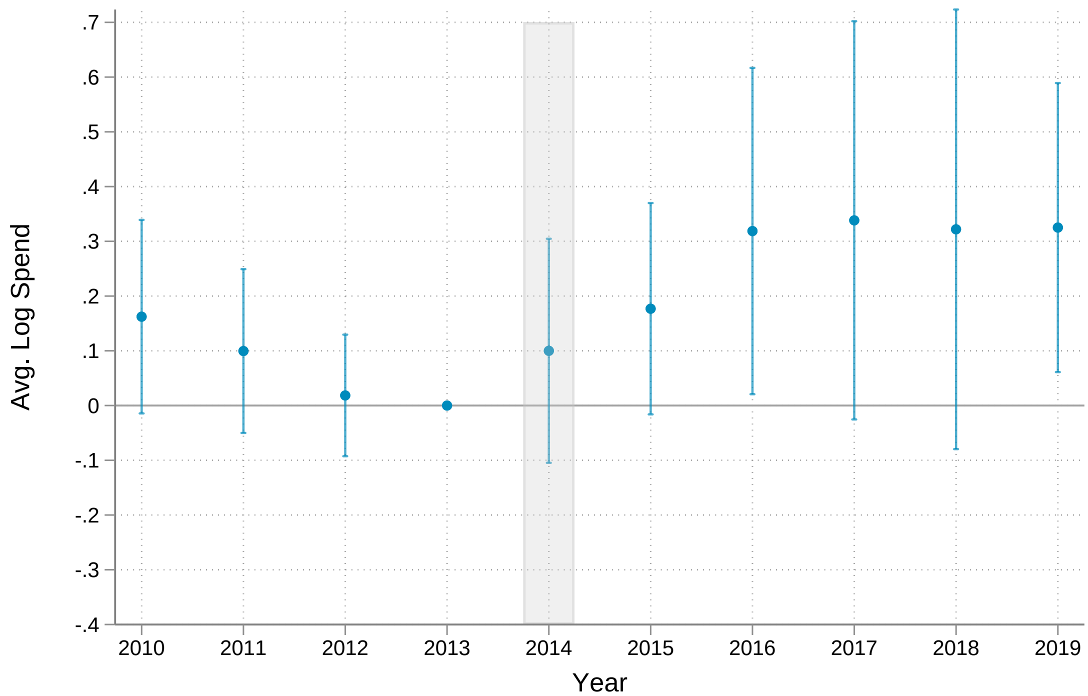
(b) Treated and control series



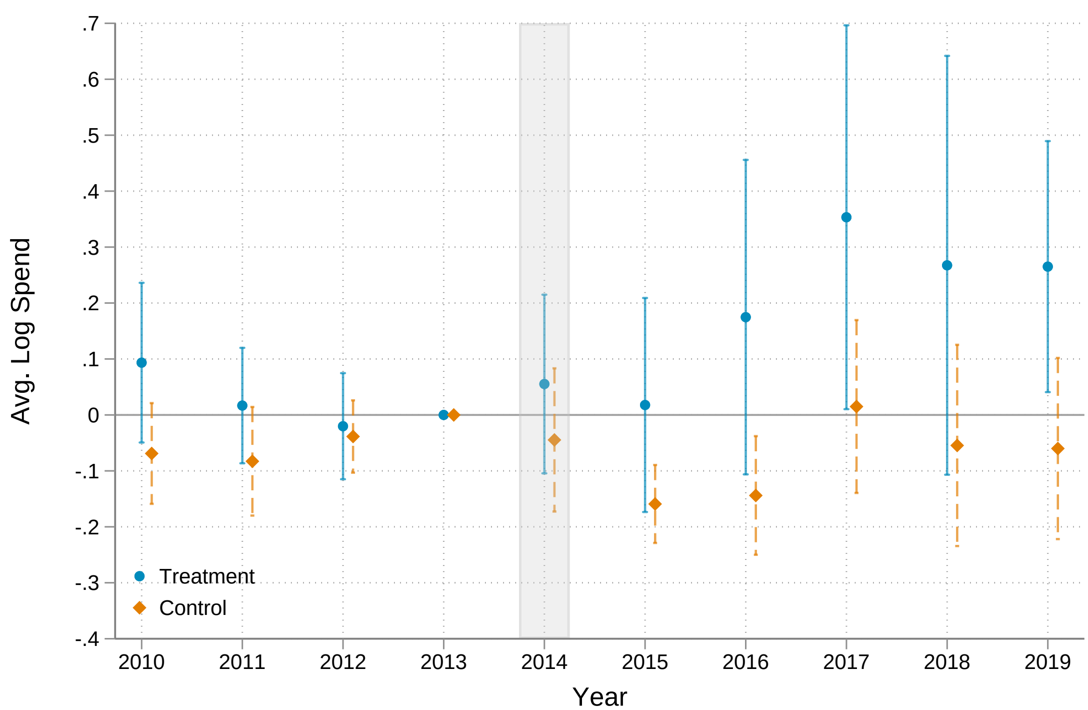
Notes: Event study estimates of average log quantity around the merger. Panel (a) plots the pooled difference between treated product markets and their matched controls. Panel (b) plots treated product markets (blue) and their matched controls (orange) separately. The vertical gray band indicates the merger year (2014). Coefficients are normalized to zero at $t = -1$. The average quantity in $t = -1$ is 17.8 purchases in treated markets and 25.7 in matched control markets.

Figure 4. Pooled Merger-Induced Spending Effects

(a) Pooled difference



(b) Treated and control series



Notes: Event study estimates of average log spending around the merger. Panel (a) plots the pooled difference between treated product markets and their matched controls. Panel (b) plots treated product markets (blue) and their matched controls (orange) separately. The vertical gray band indicates the merger year (2014). Coefficients are normalized to zero at $t = -1$. Average spending in $t = -1$ is \$2,094 in treated markets and \$1,983 in matched control markets.

Table 2. Pooled Merger Effects on Prices, Quantities, and Spending

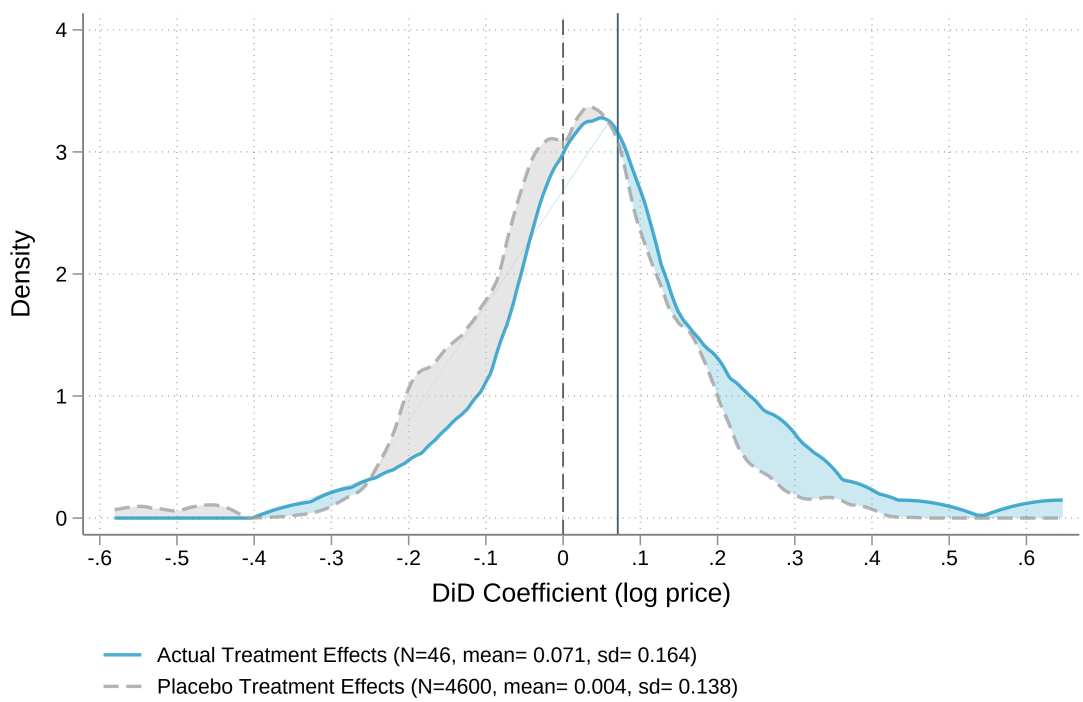
	(1)	(2)	(3)
Panel A: Log price			
Post × Treated	0.164^{*}	0.162^{*}	0.153
	(0.099)	(0.096)	(0.096)
Panel B: Log quantity			
Post × Treated	0.023	0.050	0.001
	(0.063)	(0.064)	(0.063)
Panel C: Log spending			
Post × Treated	0.199^{*}	0.225^{*}	0.167
	(0.111)	(0.121)	(0.112)
University FE	✓		
Market FE	✓		✓
Year FE	✓	✓	
University × Market FE		✓	
University × Year FE			✓
Observations	19,388	19,388	19,330

Notes: Each cell reports the pooled difference-in-differences coefficient on $D_m^T \times \text{Post}_t$ from a separate regression of the panel's log outcome on treatment, estimated on the high-confidence treated and matched-control markets, 2010–2019. Column (1) is the baseline specification with university, market, and year fixed effects. Column (2) uses university-by-market and year fixed effects. Column (3) uses university-by-year and market fixed effects. Observations are identical across panels within a column. Standard errors clustered at the market level in parentheses. These are unshrunk pooled estimates and differ from the empirical Bayes market-level price effects $\hat{\beta}_m^P$ used elsewhere in the paper. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The absence of a differential quantity response also rules out a measurement artifact in the price variable. Our records do not report unit sizes, so the 20 percent increase could in principle reflect a shift within product markets toward larger, more expensive packages, for example from a 50 mL bottle of fetal bovine serum to a 500 mL bottle, rather than a genuine rise in unit prices. Any such shift would mechanically change the recorded quantity—larger packages lowering it and smaller packages raising it—and would have to occur in treated markets relative to controls. Treated and control quantities instead move together throughout the sample, so the price increase reflects a genuine change in the unit price of a stable bundle of goods rather than a change in package size.

Behind this aggregate price effect lies substantial heterogeneity across product markets, the source of the identifying variation for our PI-level research output analysis. PIs differ in the composition of their consumables bundles, so a common set of market-level shocks generates different effects across PIs. Figure 5 plots the distribution of the shrunk market-specific price effects $\hat{\beta}_m^P$ against a placebo distribution that randomly shuffles treatment status across control markets, with the same empirical Bayes shrinkage applied to both so that the comparison reflects genuine cross-market heterogeneity rather than sampling variation. The placebo distribution is centered at zero, while the actual distribution is centered at 0.071 log points with a thicker right tail. This unweighted mean across markets is smaller than the spending-weighted pooled effect of roughly 20 percent because the pooled effect loads on the higher-spending treated markets. Both randomization inference and a Kolmogorov–Smirnov test reject the equality of our estimated treatment effects and the placebo treatment effects.

Figure 5. Market-Level Merger-Induced Price Effects



Notes: The figure overlays the unweighted distribution of market-specific difference-in-differences price effects $\hat{\beta}_m^P$ across the 46 treated markets (solid) against a placebo distribution constructed by randomly assigning treatment status across control markets (dashed). The placebo distribution pools 100 random reassignments of treated status among the control markets, each of the same size as the treated set, for 4,600 placebo estimates in total, with the same empirical Bayes shrinkage applied to both distributions. The reported p-values are from randomization inference, which compares the mean of the estimated treated effects to the placebo distribution ($p < 0.01$), and from a Kolmogorov–Smirnov test of the equality of the two distributions ($p = 0.035$).

4 The Effect of Higher Input Prices on Research Output

Higher consumables costs translate into less scientific output. To trace the cost shock from the market-level price effects of Section 3 to research output, we assemble an annual panel of PI publications and construct a PI-level exposure measure that captures the increase in the cost of each PI's pre-merger consumables bundle. More-exposed PIs publish fewer papers after the merger than less-exposed PIs, implying an elasticity of output with respect to consumables costs of about −0.42. The decline concentrates among early-career PIs and varies with career experience in a way that observable financial resources do not explain.

4.1 Data

Our research output analysis draws on two data sources. The FOIA records link spending in each market to individual PIs, and publication records from PubMed and OpenAlex form the PI-year panel of research output measures. For the PIs whose purchases the FOIA records identify, we combine their pre-merger spending with the market-level price effects estimated in Section 3 to form the exposure measure, and for the remaining PIs we impute it from the textual similarity of their publications, with the construction detailed in Section 4.2.

4.1.1 Data on PI Spending

Building a PI's exposure measure requires linking each purchase to the PI responsible for it, since exposure depends on how a PI's spending is distributed across markets. Our procurement records, however, identify only the purchasing university, not the individual PI. We recover this link from our FOIA requests, which yielded records that can be tied to individual PIs from five institutions: UT Dallas, UMich, the University of Texas at Austin (UT Austin), Texas Tech University (TTU), and East Carolina University (ECU). Beyond purchase-level spending, these universities report either the name of the responsible PI or the federal award number that funded the purchase. When only an award number is available, we identify the funded PI by searching the award through NIH RePORTER, NSF Award Search, USAspending.gov, and paper acknowledgments, attributing 347,989 of the 396,227 purchases (92% of spending) to a PI.

4.1.2 Publication Records

We measure each PI's research output from annual publication counts, using PubMed to define the universe of life sciences publications and OpenAlex to attribute each publication to an individual PI.

PubMed Medical Subject Headings (MeSH) Extract. PubMed's Medical Subject Headings (MeSH), maintained by the National Library of Medicine, provide a curated taxonomy for biomedical literature that lets us define life sciences publications precisely. Our query covers the MeSH descriptors *Anatomy*, *Chemicals and Drugs*, *Organisms*, *Phenomena and Processes*, *Diseases*, *Anesthesia and Analgesia*, *Therapeutics*, and *Clinical Study*, which jointly capture basic, translational, and clinical life sciences research.[8] The predominant MeSH terms in the resulting sample include descriptors such as *Signal Transduction, Mutations, Receptors, Cells, Gene Expression Regulation,* and *DNA*, consistent with our goal of measuring output across the spectrum of life sciences research. We restrict to journal articles and exclude comments, case reports, technical notes, letters, and reviews. Our output measure includes all journal articles in this corpus.

OpenAlex. OpenAlex provides the consistent author and institutional identifiers we need to link each publication to a PI, which PubMed does not supply. We therefore supplement PubMed with OpenAlex metadata, drawing on a comprehensive open catalog of the global research system developed by the nonprofit OurResearch (Priem et al., 2022). OpenAlex aggregates and standardizes information from sources including PubMed, Microsoft Academic Graph, Crossref, and ORCID. It disambiguates authors with an algorithm that draws on names, publication records, citation patterns, and ORCID identifiers where available, and it standardizes

[8] Our results are robust to dropping the clinical life sciences papers from our sample (not reported).

institutional affiliations by parsing raw affiliation strings with a machine learning model.

4.1.3 PI Research Output Panel

We define PIs as the last author on publications, following the life sciences publication convention where the last author is the one who directs the project. Authors often list multiple affiliations, so we assign each PI a single home institution per year from their full publication history, taking the most consistently listed affiliation and, when affiliations tie or alternate, favoring continuity with the surrounding years.[9] We restrict the panel to PIs at R1 or R2 institutions under the *Carnegie Classification*, which represent the research-intensive universities where federally funded life sciences research concentrates. We further restrict to PIs who do not change institutions during the sample period, so that each PI's fixed effect corresponds to one institution and is not confounded by moves between different research environments. The panel is balanced at the PI-year level, so a PI who publishes nothing in a year is recorded as zero publications rather than dropping out.

The resulting panel comprises 13,515 U.S. life sciences PIs observed annually from 2010 to 2019 at 198 universities, averaging 2.16 publications per PI-year before the merger. We assign each PI to one of thirty research areas by k-means clustering on publication text. The clustering is estimated on the full population of U.S. life sciences authors before we impose the panel restrictions, so the research areas are not defined by the estimation sample itself. Most PIs fall in molecular biology, biochemistry, and structural biology (43%), followed by clinical and translational medicine (28%), microbiology, virology, and infectious disease (13%), neuroscience and behavioral research (12%), and plant biology (4.1%). Table 3 reports summary statistics for the estimation panel.

[9] We keep separately branded hospitals distinct from their affiliated universities, so that Mass General Brigham, for example, is not merged into Harvard. When an author lists two affiliations in the same year, we assign the one that was primary in the previous year, or otherwise the one primary in the following year.

Table 3. Summary Statistics for the PI Research-Output Panel

	Mean	SD	Min	P25	Median	P75	Max	N
Panel A: PI-year outcomes								
Publications (last-author)	1.76	1.85	0.00	0.00	1.00	2.00	32.00	135,150
Publications (any position)	3.74	3.53	0.00	1.00	3.00	5.00	64.00	135,150
Citation-weighted publications	3.61	6.11	0.00	0.00	1.46	4.21	38.56	135,150
Coauthors per paper	4.98	3.55	0.00	3.00	4.00	6.25	95.00	102,575
Active NIH grants	1.22	1.20	0.00	0.00	1.00	2.00	12.00	75,300
NIH funding ($ thousands)	530.63	968.92	0.00	0.00	357.60	717.87	60,848.60	75,300
Panel B: PI characteristics								
Years as PI in 2014	17.0	11.4	3.0	8.0	14.0	24.0	69.0	13,515
Exposure measure	0.028	0.032	−0.191	0.009	0.021	0.046	0.219	13,515
Treated-market share S_i	0.269	0.109	0.023	0.189	0.251	0.325	0.763	13,515

Notes: The sample is the all-journal PI-year panel of R1 and R2 university PIs, 2010–2019, covering 13,515 PIs and 135,150 PI-years. Panel A reports PI-year outcomes, whose mean-median gaps reflect the skewness that motivates the Poisson and log specifications used in the main text. Publications (last-author) is the outcome in our main analysis, and publications (any position) counts a PI's papers in any authorship position. Coauthors per paper is an average over a PI's papers in a year and is observed for 102,575 PI-years, since a per-paper average requires at least one publication. Active NIH grants and annual NIH funding are observed for the 7,530 PIs matched to NIH records, 75,300 PI-years. Panel B reports PI characteristics, one observation per PI. Years as PI is the number of years since a PI's first last-author publication, measured in 2014. The exposure measure and treated-market share S_i are the shift-share cost increase and the pre-merger treated-market spending share defined in Section 4.2, observed for FOIA PIs and imputed for the rest.

4.2 Empirical Strategy

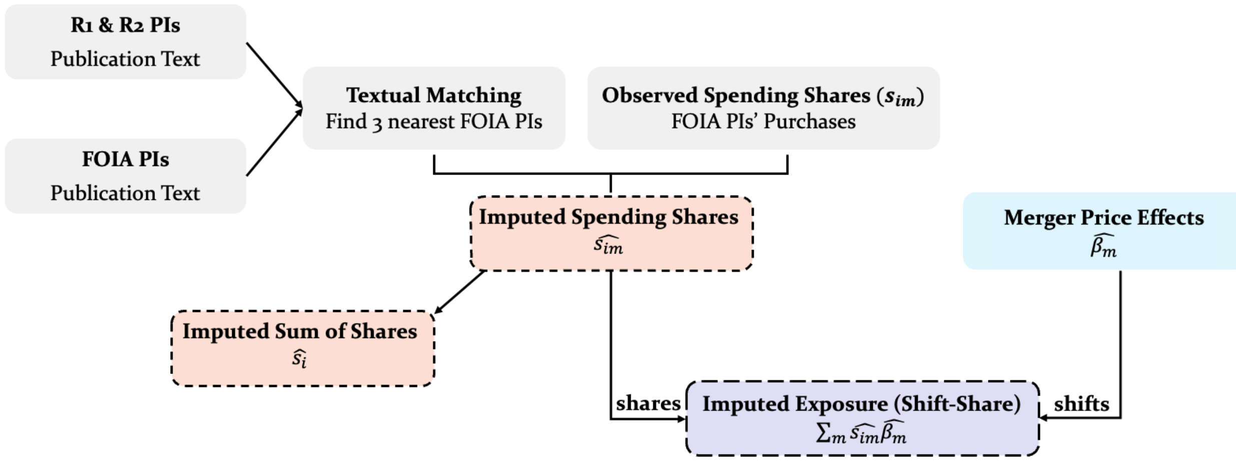
We summarize each PI's exposure to the merger as a shift-share of the market-level price effects, weighting each market's price increase by the PI's pre-merger spending in that market. A Poisson event study then traces this exposure measure to publication output, controlling for a PI's overall treated-market intensity so that identification rests on which markets a PI relied on rather than on how consumables-intensive the PI was.

Exposure measure. We observe the exposure measure directly for the 82 FOIA PIs in the panel and impute it for the remaining 13,433 as a similarity-weighted average of the exposures of their three nearest neighbors, drawn from the full set of 208 FOIA PIs whose purchases we observe and matched on the textual similarity between their pre-merger publications.[10] Figure 6

[10] Of the 208 FOIA PIs whose purchases we observe, 82 meet the panel criteria of Section 4.1, and the remainder lack last-authored publications in both the pre- and post-merger periods or publish too few papers to support estimation. All 208 nonetheless serve as imputation donors, since imputation requires only observed purchases and

summarizes the construction.

Figure 6. Construction of PI Exposure Measure as a Shift-Share



Notes: The exposure measure is a shift-share that pairs the market-level merger price effects $\hat{\beta}^P_m$ estimated in Section 3, which serve as the shifts, with each PI's pre-merger spending shares s_{im}, which serve as the shares. The shares are observed directly for the 82 FOIA PIs in the panel and imputed for the remaining 13,433 PIs by matching pre-merger publication text, namely titles, abstracts, and MeSH terms, to the three most textually similar of the 208 FOIA PIs whose purchases we observe. Weighting the shifts by the shares yields PI exposure, $\text{Exposure}_i = \sum_m s_{im}\hat{\beta}^P_m$, while summing the same shares yields $S_i = \sum_m s_{im}$, the PI's treated-market spending share, which enters the output regression as a control.

A PI's exposure measures the percentage increase in the cost of their pre-merger consumables bundle, formed by scaling each market's estimated price effect by the PI's pre-merger spending share there:

$$\text{Exposure}_i = \sum_m s_{im}\hat{\beta}^P_m \tag{2}$$

where s_{im} denotes PI i's pre-merger spending share in product market m and $\hat{\beta}^P_m$ is the empirical Bayes shrunk merger-induced price effect in market m from equation (1). We compute the spending shares relative to the PI's total pre-merger spending across the high-confidence markets defined in Section 3.2. Only treated markets have an estimated $\hat{\beta}^P_m$, so the shares that enter this shift-share measure will not sum to one but to the PI's treated-market spending share within the set of high-confidence markets. An exposure value of 0.1 corresponds to a 10 percent increase in PI i's consumables costs. Variation across PIs comes from two sources: (1) differences in the composition of consumables used (the s_{im} shares), which reflect each PI's research focus and experimental protocols, and (2) differences across markets in the size of the merger-induced

pre-merger publication text.

price effect (the $\hat{\beta}_m^P$ shifts). PIs who devoted larger pre-merger budget shares to markets that experienced larger price increases receive higher exposure values. Some PIs have a negative exposure measure, a net decrease in the cost of their pre-merger bundle, and we keep them rather than restricting to positive exposure, since conditioning the sample on the sign of the treatment would bias the estimated slope.[11]

For the FOIA PIs whose purchases we observe directly, we calculate the exposure measure from their observed pre-merger spending shares, keeping those with both pre-merger purchasing records and pre-merger publications.

For the remaining 13,433 non-FOIA PIs in the panel, we impute exposure from publication-text similarity, relying on the link between a PI's published topics and their likely consumables bundle. The underlying assumption is that PIs working on similar scientific topics use similar experimental protocols and therefore have comparable spending compositions across consumables markets. Imputed exposure is then a similarity-weighted average of the observed exposures of a PI's three most textually similar FOIA PIs, with similarity measured as the cosine similarity between TF-IDF representations of pre-merger publication text formed for each PI from titles, abstracts, and MeSH terms:

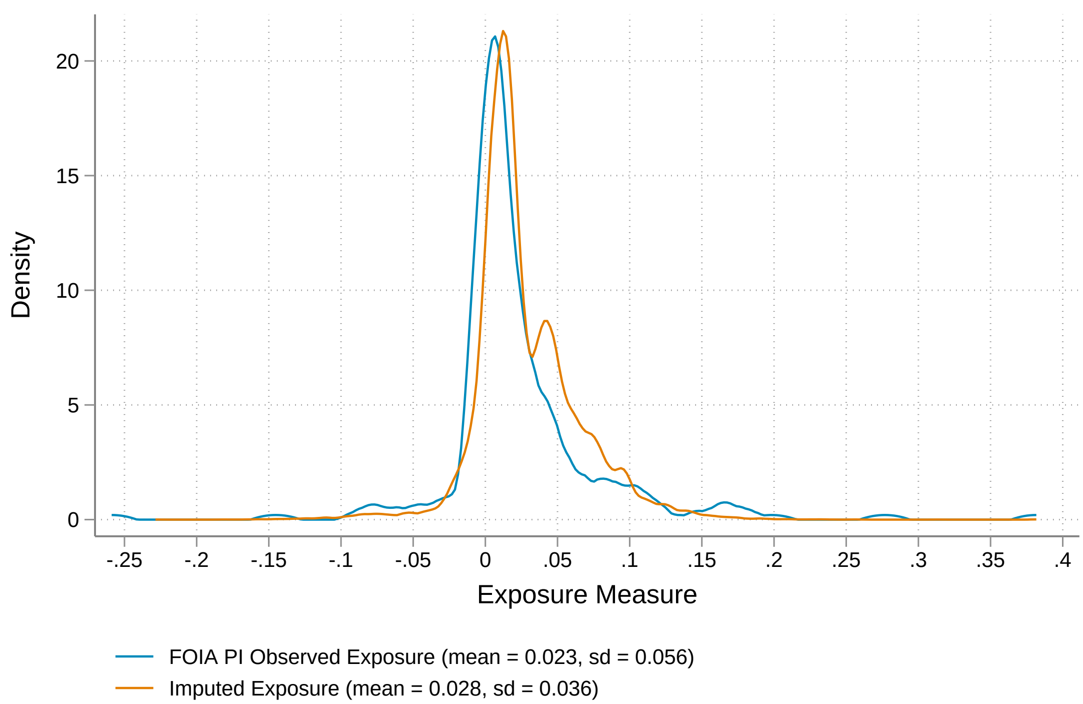
$$\widehat{\text{Exposure}}_i = \sum_{k \in \mathcal{N}_3(i)} w_{ik} \cdot \text{Exposure}_k \tag{3}$$

where $\mathcal{N}_3(i)$ denotes the set of three nearest FOIA PIs to target PI i and w_{ik} are the similarity weights normalized to sum to one. We choose three neighbors by validating the imputation on the coauthors of FOIA PIs, taking the FOIA coauthor's observed exposure measure as the benchmark for a PI whose own purchases we do not observe. For each such PI, we predict their exposure measure from their independent publication text, excluding any papers written jointly with the FOIA PI, and compare the prediction to the benchmark. Prediction accuracy peaks at three neighbors and declines on either side, which motivates the choice of $\mathcal{N}_3(i)$. Each imputed exposure measure carries a match-confidence score, the cosine similarity to the PI's nearest

[11] Table 6 reports the estimate on the positive-exposure subsample, which keeps the same sign but is less precise, since dropping the negative-exposure PIs removes identifying variation.

FOIA anchor, with the observed FOIA PIs assigned a score of one. As a check on the imputation, Section 4.3 uses this score to reestimate the output effect on the subsample of PIs with the most similar text matches, where the imputed exposure measure is most reliable. We impute exposure for the full population of U.S. life sciences authors and impose the panel restrictions of Section 4.1 only afterward, so the imputed exposure measures do not depend on the estimation sample. We restrict to PIs assigned to a research cluster that contains at least one FOIA PI, so that every panel PI works in an area where consumables purchases are observed and our imputation process rests on common support. Figure 7 compares the distribution of the imputed exposure measure across the full PI sample to the observed exposure distribution among FOIA PIs.

Figure 7. Observed and Imputed PI Exposure Distributions



Notes: The blue line shows the distribution of directly observed exposure among the 208 FOIA PIs. The orange line shows the distribution of imputed exposure for the 13,433 non-FOIA PIs in the panel.

Empirical Specification. Our empirical design treats each PI's exposure to the merger as a continuous treatment and traces the effect on research output with an event study that compares the publication trajectories of more- and less-exposed PIs around the 2014 merger. The merger generates exogenous shifts in the prices of the treated product markets, and a PI's exposure measure aggregates those shifts through the PI's pre-merger consumables bundle, so PIs who rely on the hardest-hit markets receive the largest cost shocks.

We estimate an event study of publication counts that interacts each PI's exposure measure with relative-year indicators, and the coefficients on those interactions are our parameters of interest. Our empirical specification is

$$E[y_{it} \mid \mu_i, \lambda_t] = \exp\left(\mu_i + \lambda_t + \sum_{n=-4}^{5} \beta_n \left[D_{i,t-n} \times \text{Exposure}_i\right] + \sum_{n=-4}^{5} \gamma_n \left[D_{i,t-n} \times S_i\right]\right) \tag{4}$$

where y_{it} is the raw publication count for PI i in year t. We fit it by Poisson pseudo-maximum likelihood (PPML), which is appropriate for a count outcome with many zeros and makes each β_n a semi-elasticity, the proportional effect on expected output rather than a change in the number of papers. The exposure measure is itself in proportional terms, the percent increase in the cost of a PI's pre-merger bundle, so each β_n reads as an elasticity of output with respect to consumables costs, and we interpret magnitudes with the estimates in Section 4.3. The terms μ_i and λ_t are PI and year fixed effects, $D_{i,t-n}$ are relative-year dummies centered on 2014, and Exposure_i is the imputed exposure measure constructed above. PI fixed effects absorb time-invariant differences in baseline research output, and year fixed effects capture aggregate publication trends common to all PIs. The term $S_i = \sum_m s_{im}$ is the sum of PI i's pre-merger spending shares across the treated markets, which we interact with the same relative-year indicators. We normalize the effect at $t = -1$ to zero, so that all β_n are interpreted relative to the year immediately before the merger.

The exposure measure combines two sources of variation, and our empirical design isolates one of them by controlling for the sum of shares. Since the spending shares are defined over the treated markets alone, they sum to S_i, the overall share of a PI's pre-merger budget exposed to treated markets, rather than to one. A PI's exposure is therefore high for either of two reasons. The PI may have devoted a large share of the budget to treated markets to begin with, captured by a high S_i, or, holding S_i fixed, the PI may have concentrated that budget in the treated markets whose prices rose the most. Only the second channel reflects the cost shock we study. The first channel does not, because S_i is not as good as randomly assigned. A high S_i identifies a reagent-intensive PI, and reagent intensity is correlated with field, lab scale, and baseline research trajectory, all of which affect output independently of the merger. Left uncontrolled, exposure would in part proxy for the type of PI rather than the cost shock the PI faced. Following Borusyak et al. (2022), we control for S_i directly. Holding S_i fixed, two PIs are then compared only on which treated markets they relied on and how sharply those markets were hit, which is the variation our design is meant to use. In the event study of equation (4), the PI fixed effects absorb the level of S_i, and the interactions of S_i with the relative-year indicators remove any differential trend associated with overall exposure intensity. The coefficients β_n are thereby identified from this composition-driven variation alone. Since S_i is derived from the sum of imputed market shares for the non-FOIA PIs, Appendix Figure A4 compares its imputed distribution to the observed distribution among FOIA PIs. The imputed distribution is centered close to the observed one but is more compressed, since averaging over three neighbors dampens dispersion.

We report the reduced-form effect of the exposure measure on output rather than a structural elasticity of output with respect to consumables spending. Recovering that elasticity would require relating each PI's exposure to the realized increase in their consumables costs, but PI-linked spending is observed only for the FOIA PIs, too few to estimate that relationship on the full panel with any precision. We therefore report reduced-form effects throughout and treat the implied output-cost elasticity as an interpretive benchmark rather than a structural estimand.

Our identification here relies on a common-trends assumption that, absent the merger, the publication trajectories of more- and less-exposed PIs would have evolved in parallel, conditional on the PI and year fixed effects. Our exposure measure is a shift-share aggregation of the market-level price effects, formed from spending shares fixed at their pre-merger values, so we place the identifying assumption on the exogeneity of the market-level shifts rather than of the shares (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022). This requires that the merger-induced price effects $\hat{\beta}_m^P$ are unrelated to the counterfactual research trajectories of the PIs who bought in those markets. This is plausible because the size of a market's price increase reflects the merging firms' pre-merger overlap in that market, a supply-side feature that should be unrelated to the research output of the PIs who buy those inputs. The identifying variation comes from price shifts across 46 treated markets. Some of these shifts may be correlated because markets used in the same molecular-biology workflows can experience similar price changes, reducing the effective number of independent shocks. At the same time, the merger generated substantial variation in price effects across markets, with larger increases where the merging firms' pre-merger overlap was greater.[12] This variation arose from the combination of the Thermo Fisher Scientific and Life Technologies product portfolios, rather than from where high-output PIs were concentrated or how their research was evolving. Thus, while the market-level shifts need not be fully independent, their magnitude is driven by pre-merger market structure that is plausibly orthogonal to PIs' counterfactual research trajectories.

4.3 Results

More-exposed PIs publish fewer papers after the merger. The decline concentrates among early-career PIs in a way that observable financial resources do not explain. The estimate is stable across alternative empirical specifications.

Total publication output. Exposure to the cost shock lowers publication output with an output-cost elasticity of approximately −0.42. The decline emerges after the merger and builds

[12] Consistent with this mechanism, the estimated market-level price effects correlate positively, although imprecisely, with a pre-merger Herfindahl–Hirschman index computed from our procurement records ($r = 0.156$). The supplier-name inconsistencies described in Section 3.2 make this index too noisy to define treatment, and the same noise attenuates the correlation, so we read it as directional corroboration of the market-power mechanism rather than as a measurement of concentration.

over the following years. In Panel (a) of Figure 8, the event study shows that the pre-merger coefficients are individually insignificant and show no systematic trend, while the post-merger coefficients are negative and growing over time. Pooling the pre- and post-merger years, Table 4 summarizes the Poisson difference-in-differences estimates from regressing annual publication counts on the interaction of post-merger timing with the exposure measure. Controlling for the treated-market share, which isolates the price-shift variation from overall treated-market intensity, the estimated coefficient is −0.421, compared with −0.417 without the control. Panel (b) of Figure 8 reports the corresponding binned scatterplot.

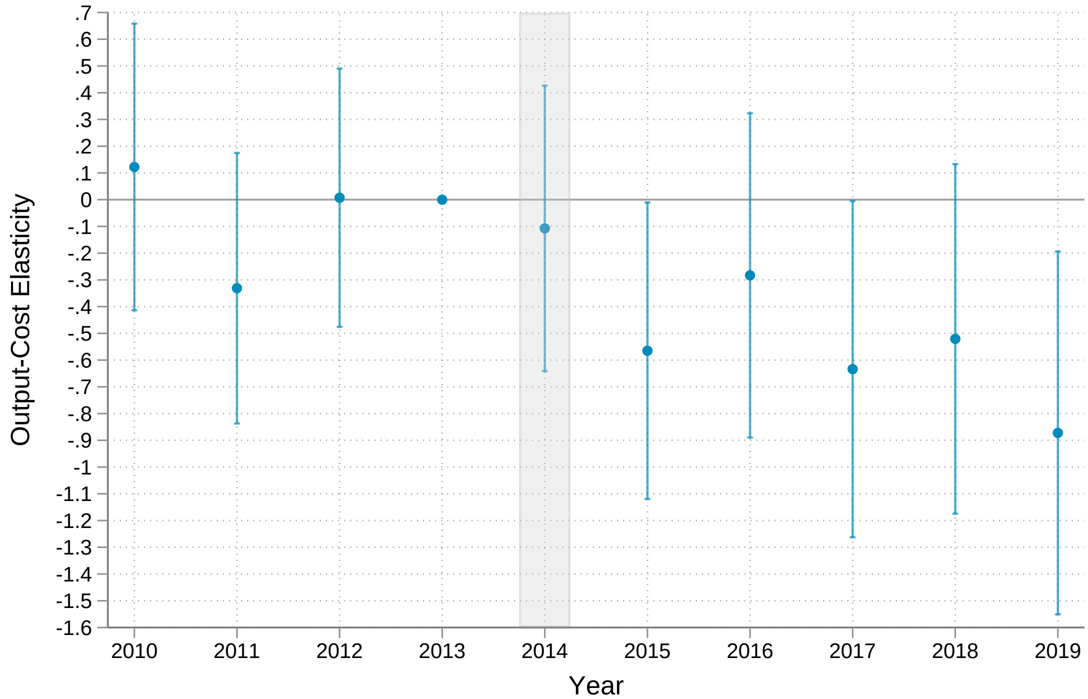
Since the exposure measure is the percent increase in the cost of a PI's pre-merger consumables bundle, the Poisson coefficient can be read as an elasticity of research output with respect to consumables cost. For the average PI, whose consumables costs rose by 2.8 percent, it implies a 1.2 percent decline in output, or roughly 0.026 fewer papers per year. This elasticity is measured against consumables costs alone, not the full grant, and should not be interpreted as a grant funding-level effect. Consumables account for only roughly 10 to 20 percent of a grant's direct costs, and rescaling to the grant level would require assuming that a dollar lost to input prices reduces output in the same way as a dollar cut from any other budget line, an assumption our design cannot support.

In dollar terms, each additional $10,500 in annual consumables costs corresponds to one fewer publication per PI per year. Figure 9 shows that within the high-confidence markets over which the exposure measure is defined, the average UT Dallas and UMich PI spends about $9,500 per year, with some spending as much as $70,000. The 2.8 percent cost increase on this base amounts to about $271 per year for the average PI, mapping to the 0.026 fewer papers per year.

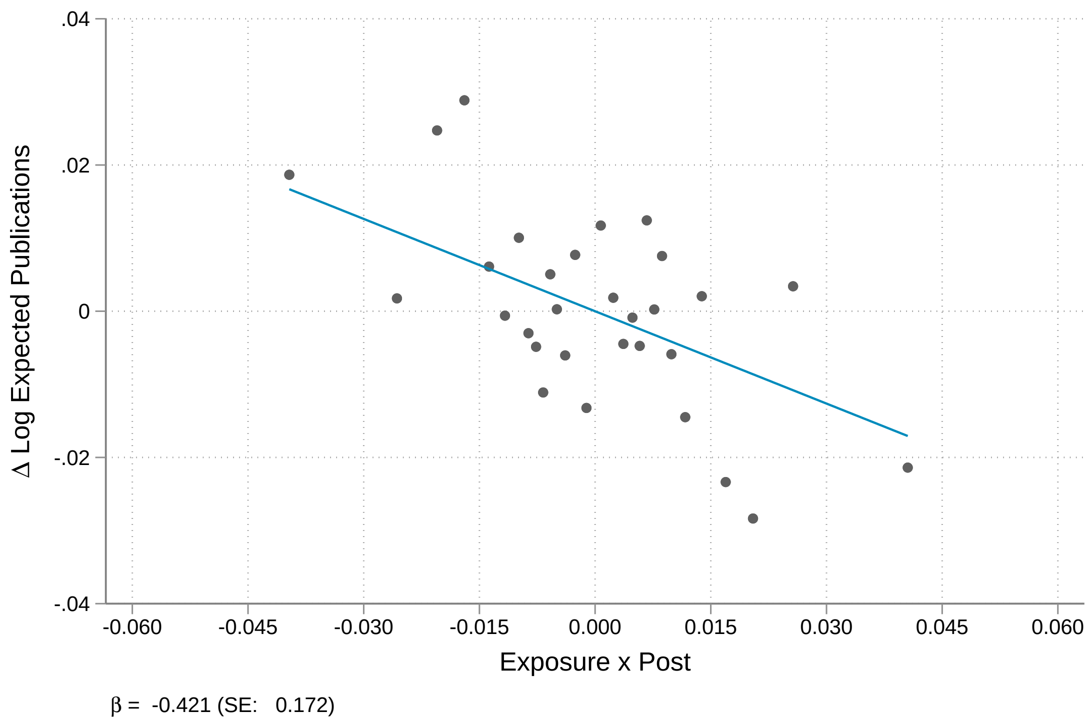
The funding margin does not appear to offset the loss. For the 7,530 panel PIs matched to NIH ExPORTER records, we show in Appendix Figure A5 that more-exposed PIs hold more active NIH research grants after the merger, and award dollars show a similar but noisier

Figure 8. Effect of the Input Cost Shock on Publication Counts

(a) Event study



(b) Difference-in-differences



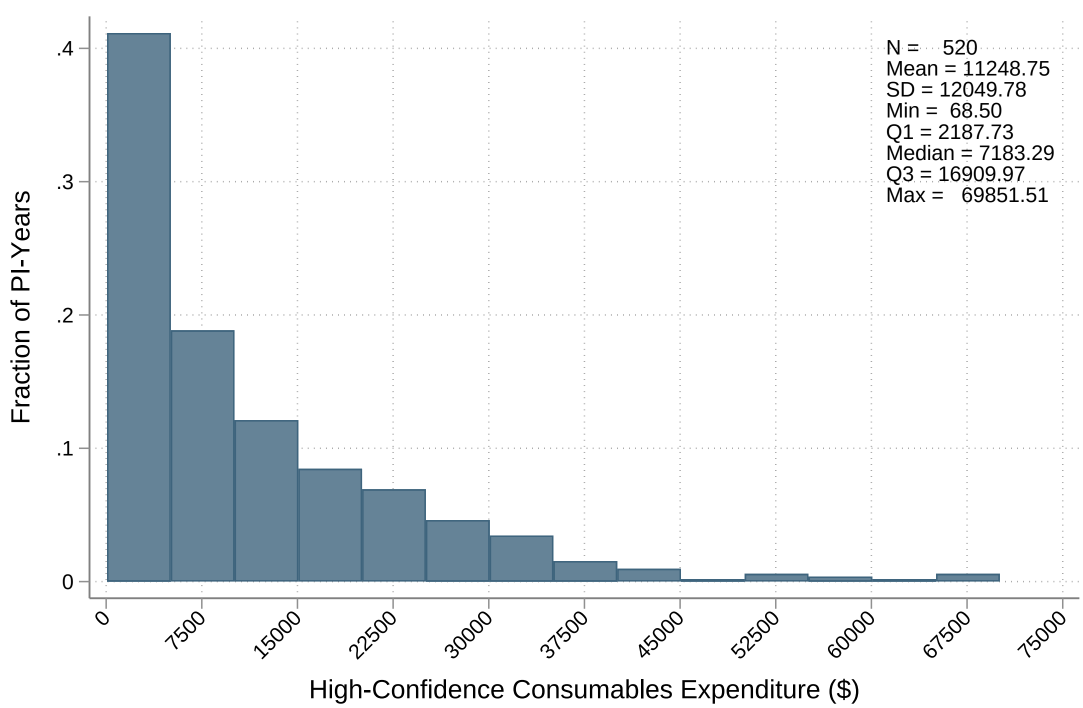
Notes: Panel (a) plots event study estimates from equation (4), the coefficients on the interaction of PI exposure with relative year indicators, with the vertical gray band indicating the merger year (2014) and coefficients normalized to zero at $t = -1$. Panel (b) is a binned scatter of annual publication counts against the $\text{Post}_t \times \text{Exposure}_i$ interaction, after partialling out PI and year fixed effects and the treated-market share control. Each point is a bin of PI-year observations. The slope of the fitted line is the difference-in-differences coefficient, $\beta = -0.421$ (SE = 0.172). The sample is the panel of 13,515 PIs at 198 universities, with a pre-merger mean of 2.16 publications per PI-year.

Table 4. Effect of the Input Cost Shock on Publication Output

	(1) With share control	(2) Without share control
Exposure measure	-0.421^{**}	-0.417^{**}
	(0.172)	(0.171)
Treated-market share S_i	0.007	
	(0.052)	
Pre-period mean	2.16	2.16
Observations	135,150	135,150

Notes: Poisson pseudo-maximum likelihood (PPML) difference-in-differences estimates of the effect of merger exposure on annual publication counts, from a panel of 13,515 PIs observed 2010–2019 with PI and year fixed effects. The reported coefficient is on $\text{Post}_t \times \text{Exposure}_i$ and is a semi-elasticity, so the implied effect is proportional to a PI's baseline output. Column (1) adds the treated-market share control following Borusyak et al. (2022) and is our preferred specification, while column (2) omits it. Standard errors clustered at the PI level in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 9. Distribution of Annual PI Consumables Spending



Notes: The figure plots the distribution of annual consumables spending per PI, computed from the manually classified UT Dallas and UMich FOIA records that link purchases to individual PIs, restricted to the high-confidence markets over which the exposure measure is defined. Averaging within PI first, spending in these markets is $9,500 per PI per year. The in-figure statistics weight each PI-year equally, which yields the higher mean shown.

pattern. Output falls even as grant activity rises, so the additional funding does not restore output within the sample window. One reading is that grant-seeking itself carries a time cost, as the hours PIs devote to preparing and securing new awards are hours taken away from research, though rising NIH budgets in the fields where more-exposed PIs work could produce a similar pattern. These estimates cover only PIs we were able to match to NIH awards, so the subsample is selected on PIs who are able to win these grants, and we treat this margin as suggestive.

As a second output measure, we reestimate equation (4) with citation-weighted output, which weights each paper by its average annual citations, and rescales the weighted counts to sum to the total number of papers, so the measure stays in units of papers and remains comparable to the raw count. The estimates are reported in Appendix Figure A6. The estimated decline is smaller than the raw-count effect but too imprecise to distinguish from either zero or the raw-count coefficient, so we draw no conclusions about the composition of the forgone papers.

Heterogeneity. The output decline appears across most types of PIs but is particularly pronounced among early-career PIs. We examine heterogeneity by reestimating the main specification across institution types and subgroups defined by the pre-merger median of PI and institutional characteristics.

The response divides most sharply by career stage, with the decline concentrated among early-career PIs. Figure 10 divides the panel at the median of 14 years since a PI's first last-authored publication as of 2014, and estimates the event study for both groups in one regression, with a separate set of exposure-by-year interactions for early- and late-career PIs. The last author in the life sciences typically directs the research project, so this measure captures the number of years a PI has spent leading an independent lab rather than simply the time since their first publication. Among early-career PIs, the decline emerges following the merger and grows through the end of the sample, while the response among late-career PIs is smaller and imprecisely estimated. In the pooled difference-in-differences version of this specification, the early-career response is -0.723 (SE 0.235) and the late-career response is -0.209 (SE 0.249). The gap between the two, -0.514 (SE 0.343), is not significant at conventional levels ($p = 0.13$).

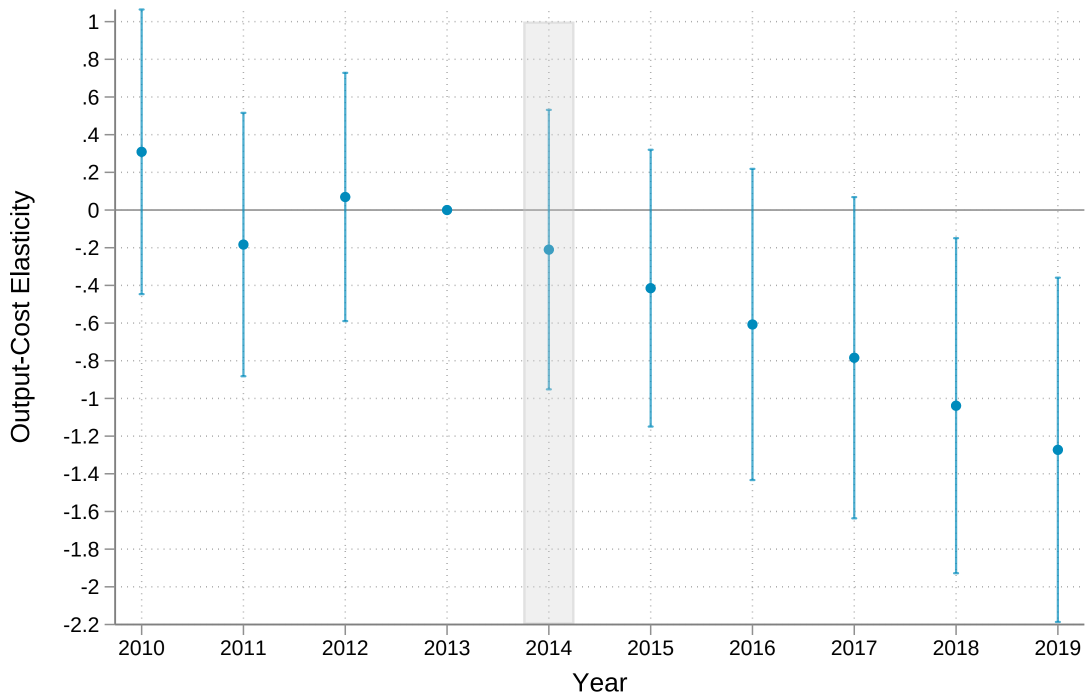
Measuring experience instead as years since a PI's first publication of any authorship, which counts years at the bench rather than years leading a lab, sharpens the split, with a gap of -0.871 (SE 0.342, $p = 0.011$). The two measures are highly correlated, and the split on bench experience, which begins to accumulate during PhD and postdoctoral training, is consistent with the tacit-knowledge reading in Section 5.2.

The larger response among early-career PIs is not mirrored by heterogeneity in observable financial resources, as Figure 11 shows. The pooled difference-in-differences estimate is negative in nearly every subgroup, but splits by baseline NIH funding, institutional research funding and endowment, and metropolitan-area size provide little evidence that better-resourced PIs or those at stronger R&D institutions experience systematically smaller declines. If anything, the point estimates run in the opposite direction, with somewhat larger declines among PIs with more baseline NIH funding and higher baseline output, though the confidence intervals overlap. These comparisons alone, however, cannot entirely rule out financial resources as an explanation for this pattern because career stage and funding are positively correlated. As Panel (a) of Figure 13 shows, before the merger early-career PIs received an average of $436,000 in annual NIH funding, compared with $633,000 among late-career PIs. The stronger response among early-career PIs could therefore partially reflect differences in funding rather than experience itself.

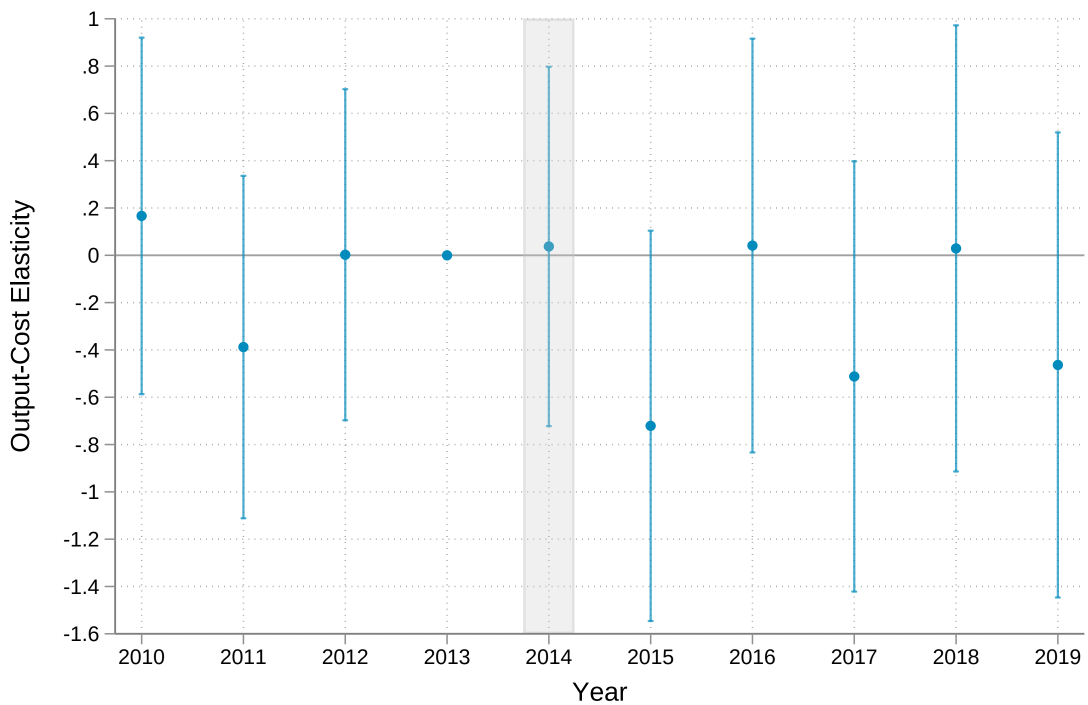
We find that the larger response among early-career PIs persists even when we compare PIs with similar baseline funding. In the NIH-matched sample, we estimate the exposure response for four groups defined by career stage and above- or below-median baseline NIH funding, reported in Figure 12. Within both funding groups, early-career PIs experience a sharper decline than late-career PIs, although the four group estimates are individually imprecise. Together with the lack of a systematic gradient across the broader resource splits in Figure 11, this

Figure 10. Effect of the Input Cost Shock on Publication Counts by Career Stage

(a) Early-career PIs



(b) Late-career PIs



Notes: The two panels plot event study estimates from a single regression that extends equation (4) with a separate set of exposure-by-relative-year interactions for each career stage, each panel showing one stage's coefficients, normalized to zero at $t = -1$, with the vertical gray band indicating the merger year (2014). The split is done at the median number of years (14) since a PI's first last-authored publication in 2014. Early-career PIs are those below the median (6,613 PIs at 195 universities, pre-merger mean 1.99 publications per year), and late-career PIs those above (6,902 PIs at 178 universities, pre-merger mean 2.32).

evidence suggests that differences in observable financial resources are unlikely to account for the substantially larger response among early-career PIs.

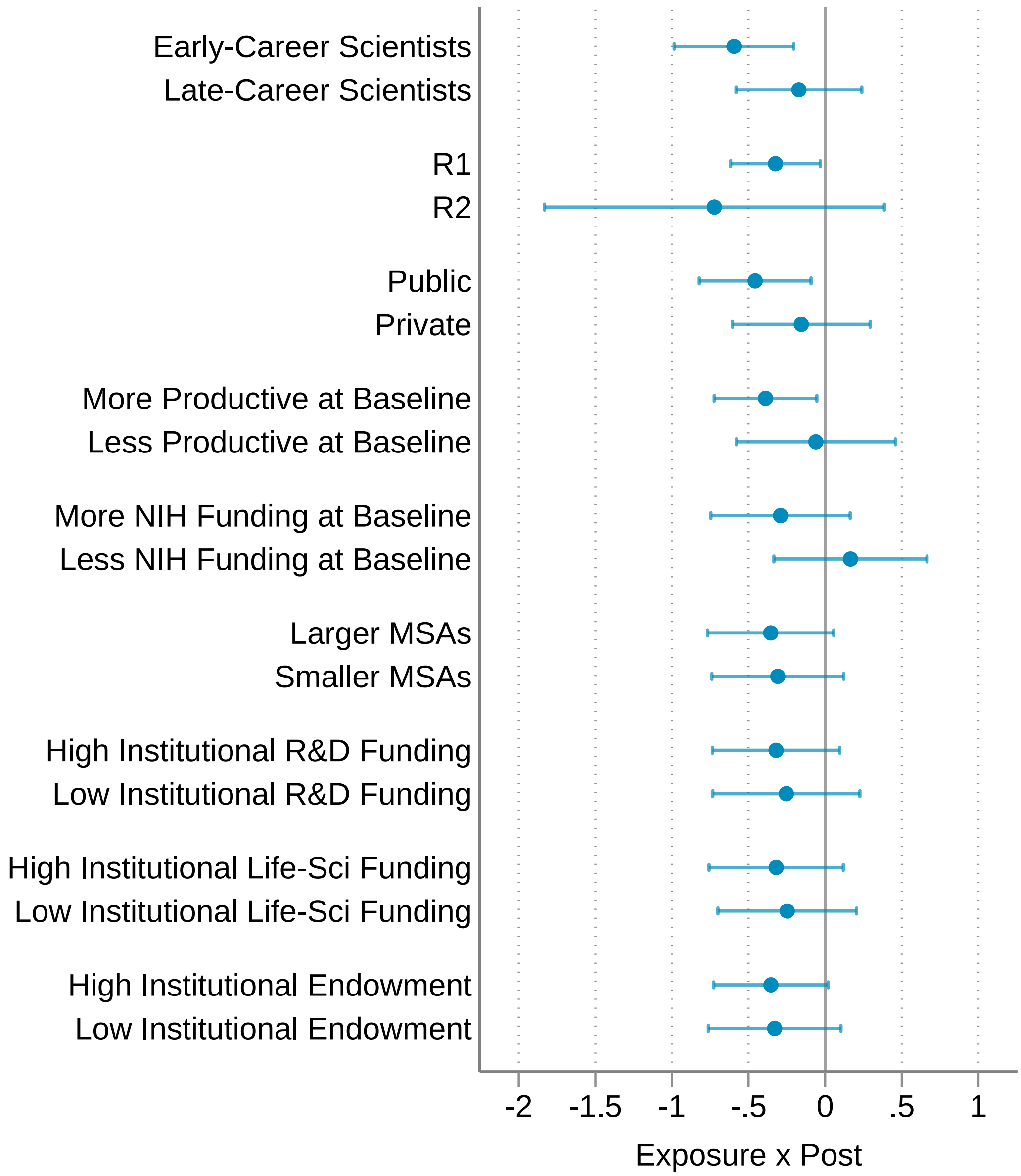
The capacity to adjust to higher consumables costs therefore appears to vary in ways that observable financial resources do not explain. One interpretation is that scientists accumulate tacit knowledge over a career, learning which inputs can be substituted, which protocols can be modified, and which projects can be delayed or redirected when inputs become more expensive. We view this interpretation as suggestive and return to it in Section 5.2, which presents supporting evidence from our survey of PIs.

Robustness. The magnitude of our reduced-form estimate is relatively stable across the functional form of the count model, the fixed-effect structure, and the construction of the exposure measure.

Table 5 reestimates the output effect under alternative functional forms and fixed-effect structures. The annual number of publications is a count outcome with mass at zero, so our preferred specification is the Poisson pseudo-maximum likelihood regression in equation (4). The odd columns estimate the model by Poisson and by ordinary least squares in levels and in the log of one plus the count, all with PI and year fixed effects, and deliver the same quantitative pattern, with implied output declines close to the baseline. The even columns replace the PI fixed effects with university and research-cluster-by-year fixed effects, controlling directly for the levels of exposure and the treated-market share, so that the comparison is cross-sectional, among more- and less-exposed PIs in the same university and the same research area in the same year. The estimates remain negative with implied declines close to the baseline, so differential post-merger trends across research fields do not appear to drive the estimated decline.

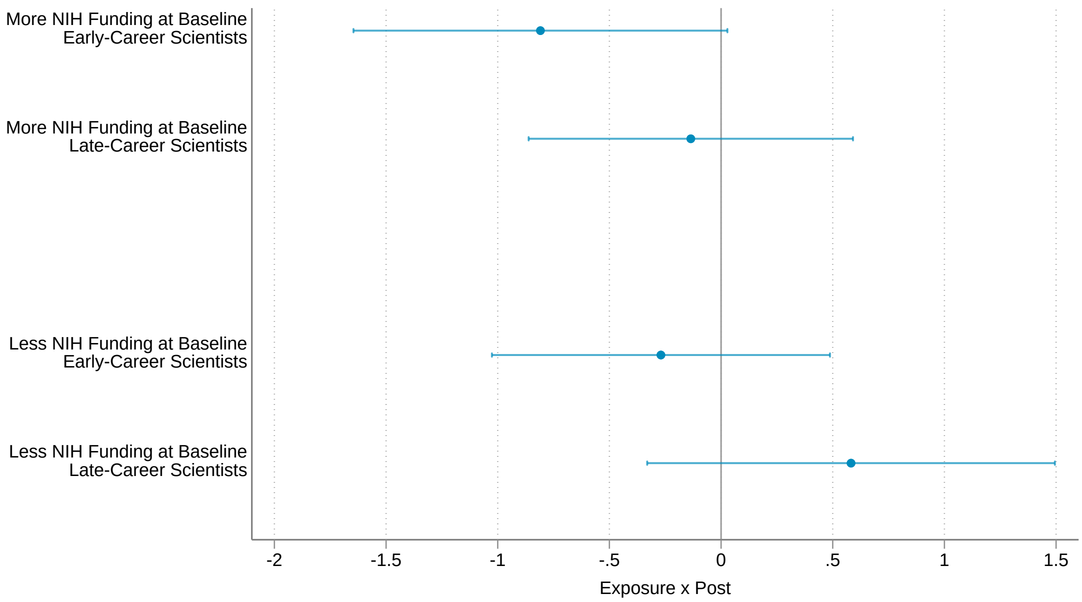
Table 6 varies the construction of the exposure measure itself. As discussed in Section 4.2, the spending shares underlying exposure sum to a PI's treated-market share rather than to one, and the baseline controls for that share directly. Renormalizing the shares to sum to one, so that exposure becomes a weighted average of the price shifts across only the treated markets a

Figure 11. Effect of the Input Cost Shock on Publication Counts by PI and Institutional Characteristics



Notes: Each row reports the Poisson difference-in-differences coefficient on $\text{Post}_t \times \text{Exposure}_i$ estimated on the panel of 13,515 PIs at 198 universities, with the two groups of each split entering one regression through separate exposure interactions. The career-stage split divides PIs at 14 years since their first last-authored publication, the panel median at 2014. The remaining rows split PIs by Carnegie classification, institutional control, and metropolitan-area size, and at the pre-merger median of baseline research output, baseline NIH funding, and institutional research and development funding, life sciences funding, and endowment. Horizontal lines are 95% confidence intervals from standard errors clustered at the PI level. Each coefficient is a semi-elasticity, a proportional effect on expected output.

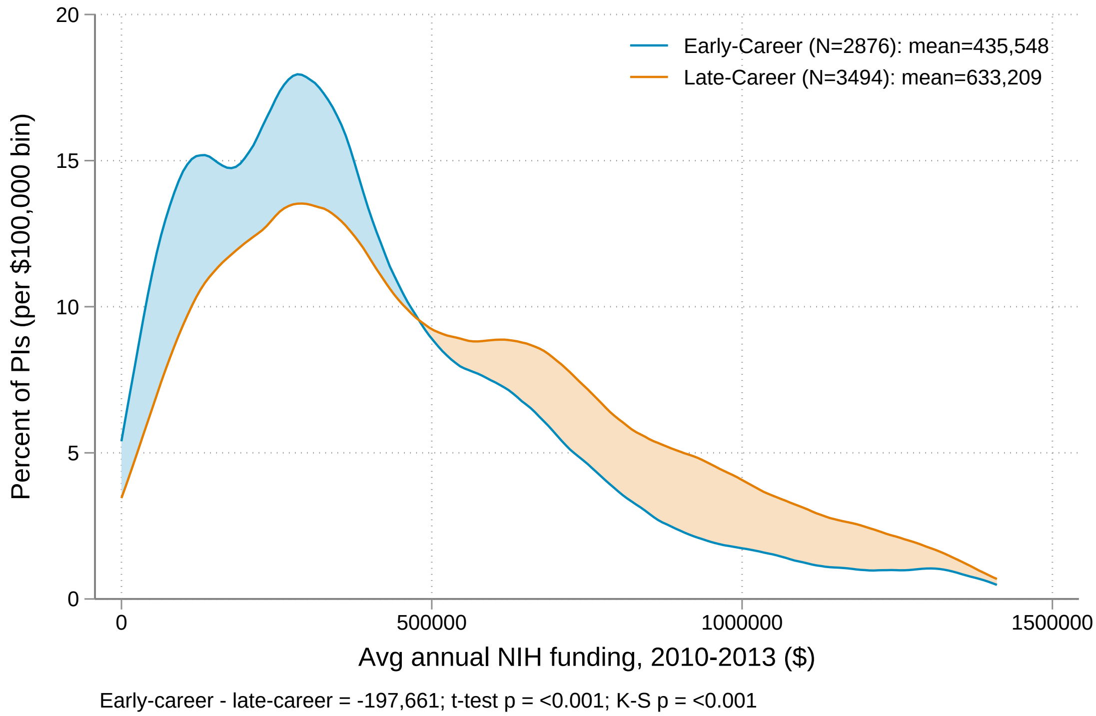
Figure 12. Exposure Response by Career Stage and Baseline NIH Funding



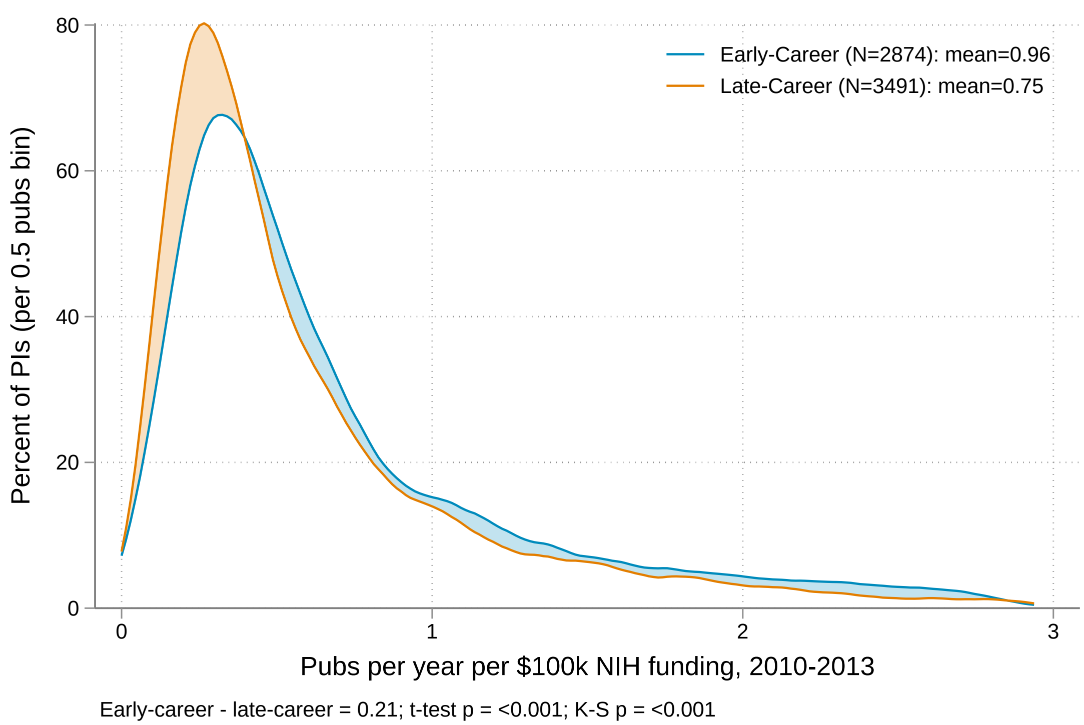
Notes: Each point is the Poisson difference-in-differences coefficient on $\text{Post}_t \times \text{Exposure}_i$ for one of the four groups defined by career stage and below- or above-median pre-merger NIH funding, estimated in a single regression on the NIH-matched PIs with a separate set of exposure and treated-market share interactions for each group, with PI and year fixed effects. Vertical lines are 95% confidence intervals from standard errors clustered at the PI level. The above-median funding groups comprise 1,444 early-career PIs at 120 universities and 2,321 late-career PIs at 114 universities, and the below-median groups 2,123 early-career PIs at 173 universities and 1,642 late-career PIs at 151 universities, together partitioning the 7,530 NIH-matched PIs.

Figure 13. Financial Resources and Output per Dollar by Career Stage

(a) Average annual NIH funding



(b) Publications per $100,000 of NIH funding



Notes: Each panel plots kernel densities separately for early-career and late-career PIs, among panel PIs with NIH funding over 2010–2013. Panel (a) is average annual NIH funding over that period and Panel (b) is publications per year per $100,000 of that funding. Early-career PIs average $436,000 in annual NIH funding against $633,000 for late-career PIs, and produce 0.96 publications per $100,000 against 0.75. Both differences are statistically significant at the 1% level.

PI uses, leaves the estimate negative with an implied output decline somewhat larger than the baseline. Restricting the sample to PIs with positive exposure reduces the identifying variation and attenuates the estimate toward zero. Weighting each PI by the text similarity of their match, so that precisely imputed exposures receive more weight, leaves the estimate negative though smaller and no longer statistically significant. Restricting to the half of the panel with the most similar matches leaves the estimate essentially at the baseline. The reduced form therefore does not appear to rest on the PIs whose exposure is imputed least reliably.

Table 5. Robustness of the Output Effect to Functional Form and Fixed Effects

	Poisson		OLS Levels		OLS Log	
	(1)	(2)	(3)	(4)	(5)	(6)
Post × Exposure						
With S_i control	-0.421^{**}	-0.415^{*}	-0.963^{***}	-1.119^{***}	-0.209^{**}	-0.195^{*}
	(0.172)	(0.213)	(0.312)	(0.352)	(0.098)	(0.107)
No S_i control	-0.417^{**}	-0.324	-1.146^{***}	-1.081^{***}	-0.244^{**}	-0.187^{*}
	(0.171)	(0.211)	(0.306)	(0.347)	(0.096)	(0.104)
Implied output decline (%)	1.2	1.2	1.3	1.5	0.9	0.8
Fixed effects						
PI + year	✓		✓		✓	
University + cluster × year		✓		✓		✓
Pre-period mean	2.16	2.16	2.16	2.16	2.16	2.16
Observations	135,150	135,150	135,150	135,150	135,150	135,150

Notes: Each column is a separate difference-in-differences regression of annual publication counts on the interaction of post-merger timing with PI exposure, reported with and without the treated-market share S_i as a control, following Borusyak et al. (2022). The outcome in the OLS Log columns is the log of one plus the count, and even columns additionally control for the levels of exposure and of S_i. The implied output decline applies the with-S_i coefficient to the average exposure of 0.028, expressed as a percent of the pre-period mean. Column (1) with the share control is the specification of Table 4, and all columns share its sample. Standard errors clustered at the PI level in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6. Robustness of the Output Effect to the Exposure Measure

	(1) Baseline	(2) Renormalized shares	(3) Positive exposure	(4) Precision-weighted	(5) High-similarity
Post × Exposure					
With S_i control	-0.421^{**}	-0.139^{***}	-0.252	-0.273	-0.419^{**}
	(0.172)	(0.053)	(0.248)	(0.177)	(0.211)
No S_i control	-0.417^{**}	—	-0.289	-0.290	-0.424^{**}
	(0.171)		(0.224)	(0.177)	(0.212)
Implied output decline (%)	1.2	1.5	0.9	0.7	1.1
Pre-period mean	2.16	2.16	2.16	2.16	2.43
Observations	135,150	135,150	120,700	135,150	67,980

Notes: As in column (1) of Table 5, each column is a Poisson (PPML) difference-in-differences regression of annual publication counts, varying the construction of the exposure measure. Column (2) renormalizes each PI's market shares to sum to one, the Borusyak et al. (2022) alternative to controlling for the treated-market share S_i, so the no-control row does not apply. Column (3) restricts the panel to PIs with strictly positive exposure. Column (4) weights each PI by the maximum text similarity of their publications to a FOIA PI, and column (5) restricts to the top 50 percent of imputed PIs by that similarity. The implied output decline applies the with-S_i coefficient (for column (2), the renormalized coefficient) to each column's average exposure, expressed as a percent of the pre-period mean. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5 Discussion

Our results show that higher input prices reduce how much research is produced. Spending in the affected markets rises with prices while quantities do not fall, and research output declines, so part of the cost shock passes through to scientific production. The capacity to adjust to the shock is uneven, with output declining the most among early-career PIs, a pattern that observable financial resources do not explain. Section 5.1 traces where the cost of higher input prices lands, and Section 5.2 asks what allows a PI to adjust to it.

5.1 The Incidence of Higher Scientific Input Prices

Our spending and output responses together trace the incidence of higher input prices as they propagate downstream to scientific production. PIs pay more for the same consumables, while scientific output declines, so the price increase costs more than the dollars that reach suppliers. Although the loss is modest for an individual PI, the same upstream suppliers serve PIs across many institutions, so the aggregate loss can be far larger.

Aggregating these effects across the 13,515 PIs in our panel quantifies the scale of this cost. For the average PI, roughly $271 per year in additional consumables spending is associated with about 0.026 fewer papers. Across our sample of PIs, these estimates imply approximately 349 fewer publications per year along with roughly $3.7 million in additional annual consumables spending. These numbers likely represent a lower bound because our exposure measure assigns a price effect only to our set of high-confidence treated markets, implicitly assigning a zero effect to the remaining treated markets. Noise in the imputed exposure measure works in the same direction, attenuating the estimated response toward zero. These aggregate calculations are meant to illustrate the macro-level scale of higher input prices rather than to estimate the merger's total welfare effect.

This output response may appear large relative to a benchmark based on total NIH grant funding, but the two align once the benchmark is scaled to the consumables share of a research budget. One fewer publication corresponds to approximately $10,500 in higher consumables costs, compared with roughly $100,000 in annual NIH funding per publication for early-career PIs and $130,000 for late-career PIs, as Panel (b) of Figure 13 shows. Part of this difference reflects the fact that NIH funding covers a broader base of spending, including personnel, equipment, and other expenses in addition to consumables. Given that consumables typically account for roughly 10 to 20 percent of research budgets, scaling the benchmark by this share implies roughly $10,000 to $27,000 per publication in consumables terms, consistent with our estimate of $10,500. This comparison therefore puts the output response in perspective but should not be interpreted as comparing the productivity of consumables with that of NIH grant funding.

The output losses fall disproportionately on PIs who produce relatively more research per dollar of NIH funding they receive. Early-career PIs, who bear the largest declines, produce more publications per dollar of public funding, as shown in Panel (b) of Figure 13, despite having only slightly lower publication rates in levels, as Appendix Figure A7 reports. They therefore sustain similar levels of scientific output with substantially less NIH funding, though this comparison is measured against NIH funding alone and does not capture start-up packages or other institutional support, which are typically concentrated early in a PI's career.

The consequences of higher input prices extend beyond publication output. Research in the life sciences drives the development of new drugs and therapeutics, so research lost today can reduce the knowledge available for future invention. Azoulay et al. (2019), for example, estimate that an additional $10 million in NIH funding generates 2.3 additional private-sector patents. We do not estimate these downstream effects, but they suggest that the publication losses we observe may capture only part of the consequences of higher scientific input prices.

5.2 What Allows a PI to Adjust to the Shock?

The larger output decline among early-career PIs points to a mechanism in which career experience helps PIs adjust, rather than to differences in financial resources. Resources could have explained this pattern, as late-career PIs may hold larger grants, work at better-resourced institutions, or otherwise have more slack to absorb higher consumables costs. Yet splits by proxies for financial resources show little evidence that better-resourced PIs experience smaller declines.

Experience can provide PIs with more ways to adjust when input costs rise. For instance, knowing where to save money and how to substitute materials, modify protocols, or alter the scale of experiments allows a PI to respond to a price increase without compromising the research itself. PIs develop this type of knowledge through years of running a lab, learning from experiments and failures. Early-career PIs have had less time to accumulate it. Under this interpretation, career stage captures accumulated knowledge of how to reorganize research practices when costs increase.

To examine this mechanism more directly, we survey active U.S. life sciences PIs about how they manage supply costs and respond to price increases. Appendix C provides additional details and the full survey instrument. Data collection is ongoing, with 19 responses to date from 255 invitations, eleven from early-career PIs and eight from late-career PIs. We classify respondents who have been running their labs for fewer than 14 years as early-career and the remainder as late-career, mirroring the career-stage cutoff in our main analysis. Given the small and selected sample, we interpret the survey as suggestive evidence on the adjustment mechanisms underlying our main results.

The clearest pattern to emerge from the survey is that cost management is learned primarily through experience rather than through formal training. The large majority of PIs report learning to manage purchasing through trial and error after starting their own lab. Those who learned from others most often name a PhD or postdoctoral advisor or a lab manager, and only two PIs report any institutional training. To the extent PIs arrive with any preparation, it comes from time spent in someone else's lab during their training years. One early-career PI reports learning by trial and error while also having "not really learned it." This suggests a learning-by-doing mechanism, in which early-career PIs face cost shocks before accumulating the knowledge to navigate them.

The survey also suggests that what experience teaches is which parts of the research itself to change, rather than a longer list of purchasing practices. One late-career PI reports having "changed [the] method of propagating virus to save costs," and another responds to rising prices by "plan[ning] more carefully the experiments." A third, when pipette tips grew scarce and expensive, considered "getting a tip washer," substituting equipment for a costly consumable. Early-career PIs describe coarser adjustments. One reports that key reagents have doubled in price since the lab opened in 2018 and that the lab is "constantly trying to scale down experiments to reduce cost." Another reports running four replicates instead of six to conserve supplies, and a third writes that "[e]verything is more expensive and we can only do less." Attempts at substitution point to the same kind of knowledge. One early-career PI "tried cheaper products, but had to switch back to the original because the cheaper ones did not work as well," and another cannot switch certain inputs "for fear of ruining the experiment." Knowing which changes preserve an experiment and which compromise it is the kind of knowledge that plausibly takes years of running a lab to accumulate.

Our survey provides suggestive evidence on how experience can explain the differential output decline we estimate for early-career PIs. Although the small, self-selected sample cannot establish that accumulated knowledge drives our results, the responses suggest that experience provides adjustment capabilities not captured by observable financial resources. In this respect, scientific research resembles other production settings in which experience and managerial capability shape responses to shocks (Reagans et al., 2005; Lee and Van den Steen, 2010; Lamorgese et al., 2024).

6 Conclusion

How much science is produced depends partly on the firms that supply it. This paper leverages the 2014 merger of Thermo Fisher Scientific and Life Technologies to study how input costs affect the production of science. Using purchase-level procurement records, we show that prices rose in markets where the merging firms had competitive overlap, while quantities purchased in those markets changed little. PIs more exposed to these price increases subsequently published fewer papers, with the decline concentrated among early-career PIs.

These findings quantify how supplier markets shape how much science public funding yields. The money is spent either way, and what changes is how much science it produces. Across the PIs in our sample, a single merger translates into approximately $3.7 million in additional annual consumables spending and roughly 349 fewer publications, from an average cost increase of 2.8 percent.

What appears to separate PIs here is less what they can spend than what they have learned to do. Our results and survey evidence suggest that adjusting to a cost shock depends partly on tacit knowledge accumulated through experience rather than on observable funding measures. This knowledge of how to reorganize projects, identify substitutes, and adjust research plans when constraints arise is acquired largely through learning on the job.

The results also highlight the purchasing environment as a relevant consideration in the design and administration of public research support. The NIH already tracks changes in the overall cost of conducting biomedical research through the Biomedical Research and Development Price Index, which informs estimates of the funding required to maintain the purchasing power of research expenditures (National Institutes of Health, 2025). An aggregate index captures price changes across a broad bundle of research inputs, so it may not reflect substantial increases within particular input markets. A sharp price increase can therefore impose significant costs on scientists who depend on a specific input without meaningfully moving the overall index. The structure of individual supplier markets is an important complement to such an index. In particular, the consequences of input price increases may depend on whether scientists can compare prices across suppliers and identify substitute products, as well as whether institutions can coordinate purchasing to reduce exposure to supplier-specific price increases.

Several open questions remain for future work. Publications capture only one dimension of how scientists may respond to higher research costs. Price increases may also change which projects scientists pursue, shift research toward less resource-intensive questions, or alter the novelty and direction of scientific work. The larger response among early-career PIs also raises a separate set of questions about the longer-run consequences of these shocks. If early-career PIs are less able to adjust, even temporary cost increases could affect subsequent funding, career progression, or retention in science. Finally, the apparent role of experience raises the question of whether the knowledge that helps scientists adjust can be better transmitted across scientists rather than acquired only through years of running a lab. Understanding these responses is important for assessing the full consequences of scientific input prices for both the production of knowledge and the scientists who produce it.

References

Arrow, Kenneth J., "Economic Welfare and the Allocation of Resources for Invention," in "Economic Welfare and the Allocation of Resources for Invention," Princeton University Press, 1962, pp. 609–626.

Ashenfelter, Orley, Daniel Hosken, and Matthew Weinberg, "Did Robert Bork Understate the Competitive Impact of Mergers? Evidence from Consummated Mergers," *The Journal of Law and Economics*, August 2014, *57* (S3), S67–S100.

Avantor, Inc., "FORM 10-K," February 2026.

Azoulay, Pierre, Joshua S. Graff Zivin, and Gustavo Manso, "Incentives and creativity: evidence from the academic life sciences," *The RAND Journal of Economics*, 2011, *42* (3), 527–554. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1756-2171.2011.00140.x.

_ , _ , Danielle Li, and Bhaven N. Sampat, "Public R&D investments and private-sector patenting: evidence from NIH funding rules," *The Review of economic studies*, 2019, *86* (1), 117–152. ISBN: 0034-6527.

_ , Wesley H. Greenblatt, and Misty L. Heggeness, "Long-term effects from early exposure to research: Evidence from the NIH "Yellow Berets"," *Research Policy*, November 2021, *50* (9), 104332.

Baker, Monya, "1,500 scientists lift the lid on reproducibility," *Nature*, May 2016, *533* (7604), 452–454.

Baruffaldi, Stefano and Fabian Gaessler, "The Returns to Physical Capital in Knowledge Production: Evidence from Lab Disasters," August 2021.

Berkrot, Bill, "Sigma-Aldrich's $17 billion buyout seen as boost for lab supplier VWR," *Reuters*, September 2014.

Bloom, Nicholas and John Van Reenen, "Measuring and Explaining Management Practices Across Firms and Countries," *The Quarterly Journal of Economics*, November 2007, *122* (4), 1351–1408.

_ , Benn Eifert, Aprajit Mahajan, David McKenzie, and John Roberts, "Does Management Matter? Evidence from India," *The Quarterly Journal of Economics*, February 2013, *128* (1), 1–51.

Borusyak, Kirill, Peter Hull, and Xavier Jaravel, "Quasi-Experimental Shift-Share Research Designs," *The Review of Economic Studies*, January 2022, *89* (1), 181–213.

Bradbury, Andrew and Andreas Plückthun, "Reproducibility: Standardize antibodies used in research," *Nature*, February 2015, *518* (7537), 27–29.

Brown, Zach Y., "Equilibrium Effects of Health Care Price Information," *The Review of Economics and Statistics*, October 2019, *101* (4), 699–712.

Cabral, Marika, Michael Geruso, and Neale Mahoney, "Do Larger Health Insurance Subsidies Benefit Patients or Producers? Evidence from Medicare Advantage," *American Economic Review*, 2018, *108* (8), 2048–87. ISBN: 0002-8282.

Caplin, Andrew, Minjoon Lee, Søren Leth-Petersen, and Matthew D. Shapiro, "How Worker Productivity and Wages Grow with Tenure and Experience: The Firm Perspective," *American Economic Journal: Macroeconomics*, 2026.

Carnehl, Christoph, Marco Ottaviani, and Justus Preusser, "Designing Scientific Grants," *Entrepreneurship and Innovation Policy and the Economy*, January 2025, *4*, 139–178.

Commission of the European Communities, "Invitrogen/ Applied Biosystems," November 2008.

Coppola, Marissa, Tessa O. House, and Jessica W. Tsai, "Maximizing grant funding: strategies for negotiating laboratory pricing," *Trends in Biochemical Sciences*, April 2025, *50* (4), 281–284.

Dafny, Leemore, Mark Duggan, and Subramaniam Ramanarayanan, "Paying a Premium on Your Premium? Consolidation in the Us Health Insurance Industry," *American Economic Review*, 2012, *102* (2), 1161–1185. ISBN: 0002-8282.

Dong, Xinwei, Dean Hyslop, and Daiji Kawaguchi, "Skill, Productivity, and Wages: Direct Evidence from a Temporary Help Agency," *Journal of Labor Economics*, April 2024, *42* (S1), S133–S181.

Donna Kridelbaugh, "Tips and Resources for Maximizing Your Lab Spend," May 2018.

Duggan, Mark, Amanda Starc, and Boris Vabson, "Who Benefits When the Government Pays More? Pass-Through in the Medicare Advantage Program," *Journal of Public Economics*, September 2016, *141*, 50–67.

Ehrenberg, Ronald G., Michael J. Rizzo, and George H. Jakubson, "Who Bears the Growing Cost of Science at Universities?," April 2003.

Ellison, Sara Fisher and Christopher M Snyder, "COUNTERVAILING POWER IN WHOLESALE PHARMACEUTICALS," 2010.

Federal Trade Commission, "In the Matter of Thermo Fisher Scientific Inc., a corporation.," January 2014.

Freedman, Leonard P., Iain M. Cockburn, and Timothy S. Simcoe, "The Economics of Reproducibility in Preclinical Research," *PLOS Biology*, June 2015, *13* (6), e1002165.

Furman, Jeffrey L. and Florenta Teodoridis, "Automation, Research Technology, and Researchers' Trajectories: Evidence from Computer Science and Electrical Engineering," *Organization Science*, March 2020, *31* (2), 330–354.

— **and Scott Stern**, "Climbing atop the Shoulders of Giants: The Impact of Institutions on Cumulative Research," *American Economic Review*, August 2011, *101* (5), 1933–1963.

Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift, "Bartik Instruments: What, When, Why, and How," *American Economic Review*, August 2020, *110* (8), 2586–2624.

Grennan, Matthew, "Price Discrimination and Bargaining: Empirical Evidence from Medical Devices," *American Economic Review*, 2013, *103* (1), 145–177. ISBN: 0002-8282.

— **and Ashley Swanson**, "Transparency and Negotiated Prices: The Value of Information in Hospital-Supplier Bargaining," *Journal of Political Economy*, April 2020, *128* (4), 1234–1268.

Hendel, Igal and Yossi Spiegel, "Small Steps for Workers, a Giant Leap for Productivity," *American Economic Journal: Applied Economics*, January 2014, *6* (1), 73–90.

Howell, Sabrina T., "Financing Innovation: Evidence from R&D Grants," *American Economic Review*, April 2017, *107* (4), 1136–1164.

Innovative Bioscience, "Where the consumables budget actually goes."

Jacob, Brian A. and Lars Lefgren, "The impact of research grant funding on scientific productivity," *Journal of Public Economics*, October 2011, *95* (9), 1168–1177.

Lamorgese, Andrea, Megha Patnaik, Andrea Linarello, and Fabiano Schivardi, "Management Practices and Resilience to Shocks: Evidence from COVID-19," *Management Science*, December 2024, *70* (12), 9058–9072.

Lee, Deishin and Eric Van den Steen, "Managing Know-How," *Management Science*, February 2010, *56* (2), 270–285.

Li, Danielle, Pierre Azoulay, and Bhaven N. Sampat, "The applied value of public investments in biomedical research," *Science*, 2017, *356* (6333), 78–81. ISBN: 0036-8075.

Lorsch, Jon R., Francis S. Collins, and Jennifer Lippincott-Schwartz, "Fixing problems with cell lines," *Science*, December 2014, *346* (6216), 1452–1453.

McKeon, John and Justine Boudou, "Innovation Under Resource Constraints: Supercomputing in Scientific Research," June 2024.

Merck KGaA, "Annual Report 2025," 2025.

Murray, Fiona, Philippe Aghion, Mathias Dewatripont, Julian Kolev, and Scott Stern, "Of Mice and Academics: Examining the Effect of Openness on Innovation," *American Economic Journal: Economic Policy*, February 2016, *8* (1), 212–252.

Myers, Kyle, "The Elasticity of Science," *American Economic Journal: Applied Economics*, October 2020, *12* (4), 103–134.

National Institute of Allergy and Infectious Diseases, "Create a Budget," January 2026.

National Institutes of Health, "Price Indexes," 2025.

Nelson, Richard R., "The Simple Economics of Basic Scientific Research," *Journal of Political Economy*, 1959, *67* (3), 297–306.

Papay, John P. and Matthew A. Kraft, "Productivity returns to experience in the teacher labor market: Methodological challenges and new evidence on long-term career improvement," *Journal of Public Economics*, October 2015, *130*, 105–119.

Priem, Jason, Heather Piwowar, and Richard Orr, "OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts," June 2022. arXiv:2205.01833 [cs].

Reagans, Ray, Linda Argote, and Daria Brooks, "Individual Experience and Experience Working Together: Predicting Learning Rates from Knowing Who Knows What and Knowing How to Work Together," *Management Science*, June 2005, *51* (6), 869–881.

Rehman, Sarah, "Technology Digest: the Importance of Reagent Consistency in Culturing Cells for Therapeutic Applications," *Regenerative Medicine*, July 2022, *17* (7), 427–429. _eprint: https://doi.org/10.2217/rme-2022-0070.

Stephan, Paula, *How Economics Shapes Science*, Harvard University Press, 2012.

Tham, Wei Yang, "Science, interrupted: Funding delays reduce research activity but having more grants helps," *PLOS ONE*, April 2023, *18* (4), e0280576.

Thermo Fisher Scientific Inc., "FORM 10-K," February 2026.

Torrington, Ebony, "Reproducibility, Scientific Rigor and Responsibility," *BioTechniques*, August 2021, *71* (2), 400–402. _eprint: https://doi.org/10.2144/btn-2021-0066.

Waldinger, Fabian, "Bombs, Brains, and Science: The Role of Human and Physical Capital for the Creation of Scientific Knowledge," *The Review of Economics and Statistics*, December 2016, *98* (5), 811–831.

Walsh, John P., Charlene Cho, and Wesley M. Cohen, "View from the Bench: Patents and Material Transfers," *Science*, September 2005, *309* (5743), 2002–2003.

_ **, Wesley M. Cohen, and Charlene Cho**, "Where excludability matters: Material versus intellectual property in academic biomedical research," *Research Policy*, October 2007, *36* (8), 1184–1203.

Weber, Tilo, Joachim Wiest, Stina Oredsson, and Karen Bieback, "Case Studies Exemplifying the Transition to Animal Component-free Cell Culture," *Alternatives to Laboratory Animals*, September 2022, *50* (5), 330–338.

A Appendix Tables and Figures

Table A1. Top Universities, Suppliers, and Product Markets by Consumables Spending

	Total Spending ($)	% of Total
Panel A. Universities		
University of Florida	27,665,553	24.3%
University of Michigan at Ann Arbor	14,979,061	13.1%
Georgia Institute of Technology	11,675,414	10.2%
University of Kentucky	6,477,970	5.7%
Wayne State University	5,582,580	4.9%
Texas A&M University	5,023,128	4.4%
Texas Tech University	4,901,626	4.3%
East Carolina University	4,678,600	4.1%
Florida State University	4,449,392	3.9%
Washington State University	3,958,014	3.5%
Panel B. Suppliers		
Thermo Fisher Scientific	47,020,957	41.3%
VWR	16,746,159	14.7%
Life Technologies	7,411,070	6.5%
MilliporeSigma	7,356,622	6.5%
Abcam	3,269,129	2.9%
Cell Signaling Technology	2,413,615	2.1%
Bio-Rad Laboratories	2,099,210	1.8%
Qiagen	1,946,813	1.7%
USA Scientific	1,510,083	1.3%
Santa Cruz Biotechnology	1,384,289	1.2%
Panel C. Product Markets		
Primary Antibodies	16,540,087	14.5%
Pipette Tips	6,319,587	5.5%
U.S.-origin fetal bovine serum	5,810,406	5.1%
Nitrile Gloves	5,214,122	4.6%
Serological Pipettes	4,763,075	4.2%
Vials	3,452,481	3.0%
Synthetic DNA Oligonucleotide – Desalted	2,924,777	2.6%
Cell Culture Plates	2,596,718	2.3%
Petri Dishes	2,348,829	2.1%
Secondary Antibodies	2,015,528	1.8%

Notes: The ten universities, suppliers, and product markets with the highest total consumables spending across the high-confidence product markets, summed over all purchases from 2010 to 2019 before the 12 control markets with contemporaneous shocks are removed from the estimation sample, with shares relative to total spending in this descriptive sample. Thermo Fisher Scientific and Life Technologies are the two merging firms. Several of the highest-spending markets, including primary antibodies, nitrile gloves, and synthetic DNA oligonucleotides, are among the markets removed because they experienced their own contemporaneous shocks, as detailed in Section 3.2.

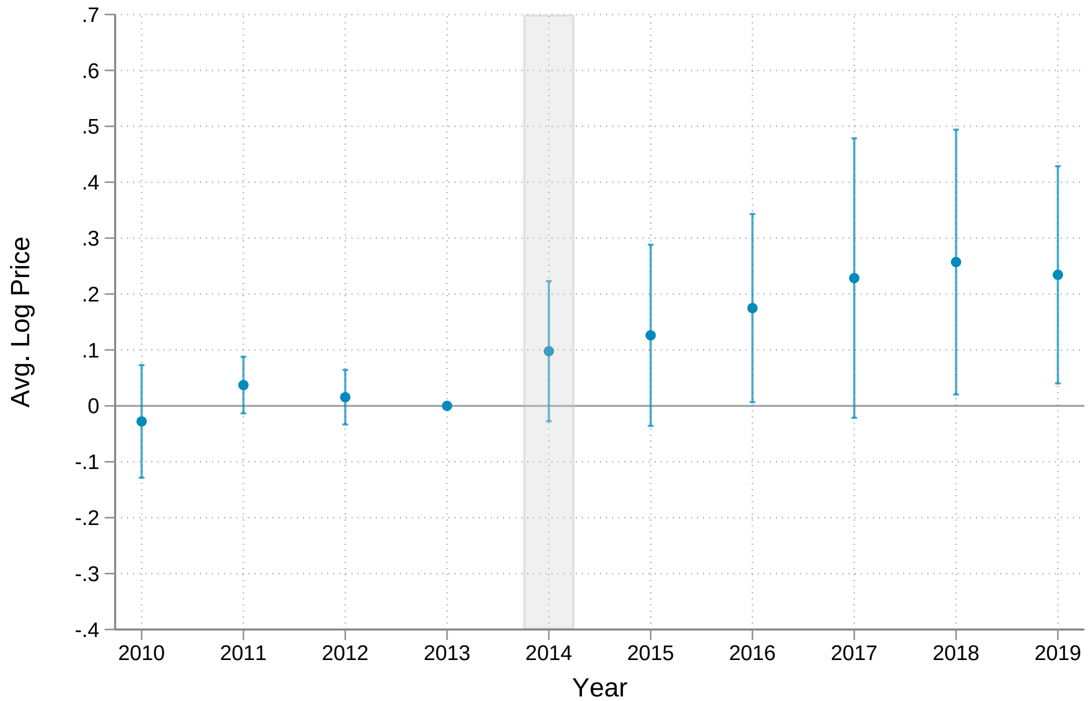
Table A2. Placebo-Timing Tests for the Merger Difference-in-Differences

	Placebo 2012		Merger 2014	
	$\hat{\beta}$	SE	$\hat{\beta}$	SE
Average log price	–0.034	(0.032)	0.164*	(0.099)
Log quantity	–0.070	(0.064)	0.023	(0.063)
Log spending	–0.109*	(0.064)	0.199*	(0.111)

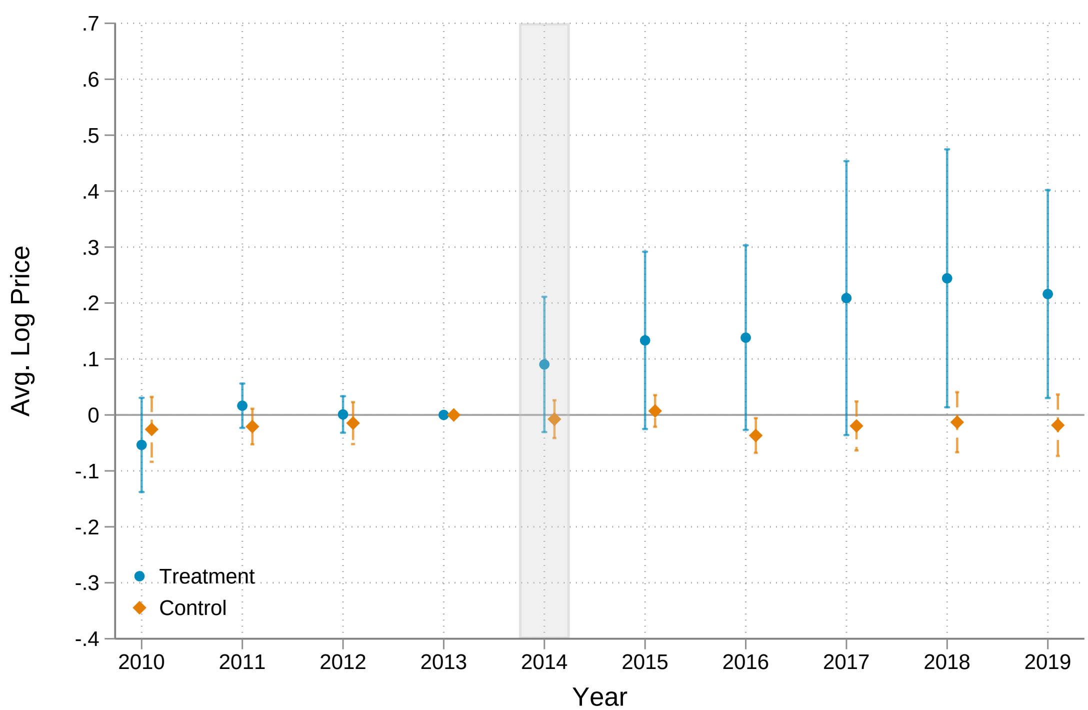
Notes: The sample is the matched treated and control markets pooled across universities. The placebo columns estimate the pooled difference-in-differences on the pre-merger panel, 2010–2013, with a false post indicator for years $t \geq 2012$, and the merger columns report the baseline estimates on the full 2010–2019 panel with the post indicator for years $t \geq 2014$. Both specifications are weighted by 2013 spending and include year, university, and market fixed effects, with standard errors clustered at the market level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A1. Pooled Merger-Induced Price Effects, Naive Specification

(a) Pooled difference



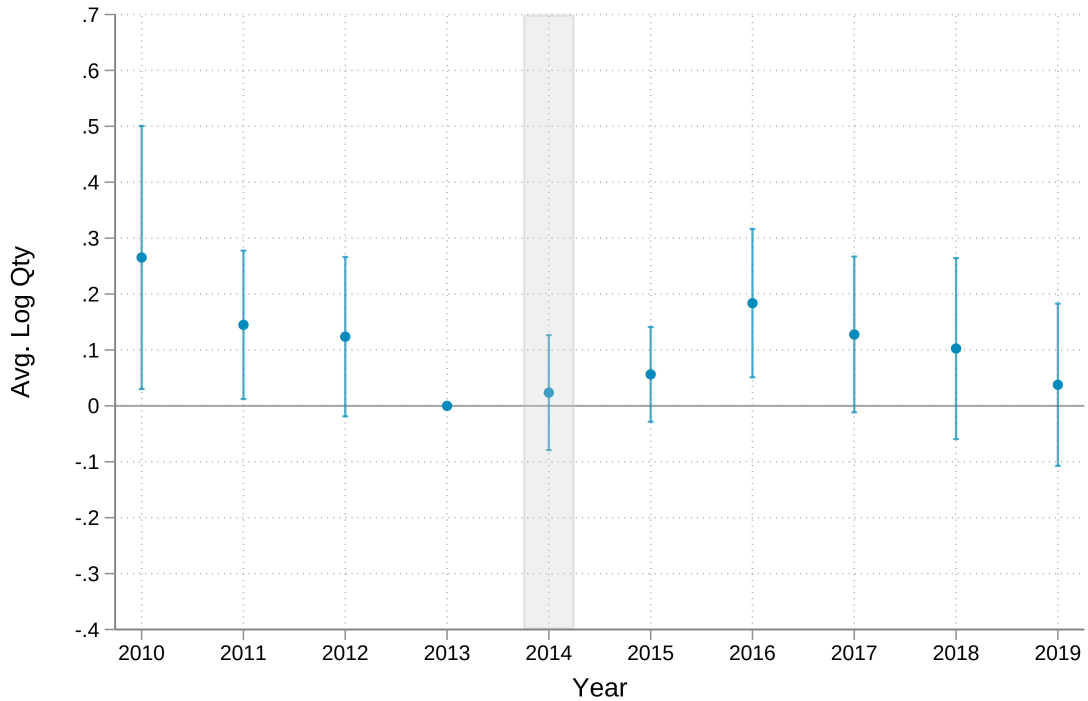
(b) Treated and control series



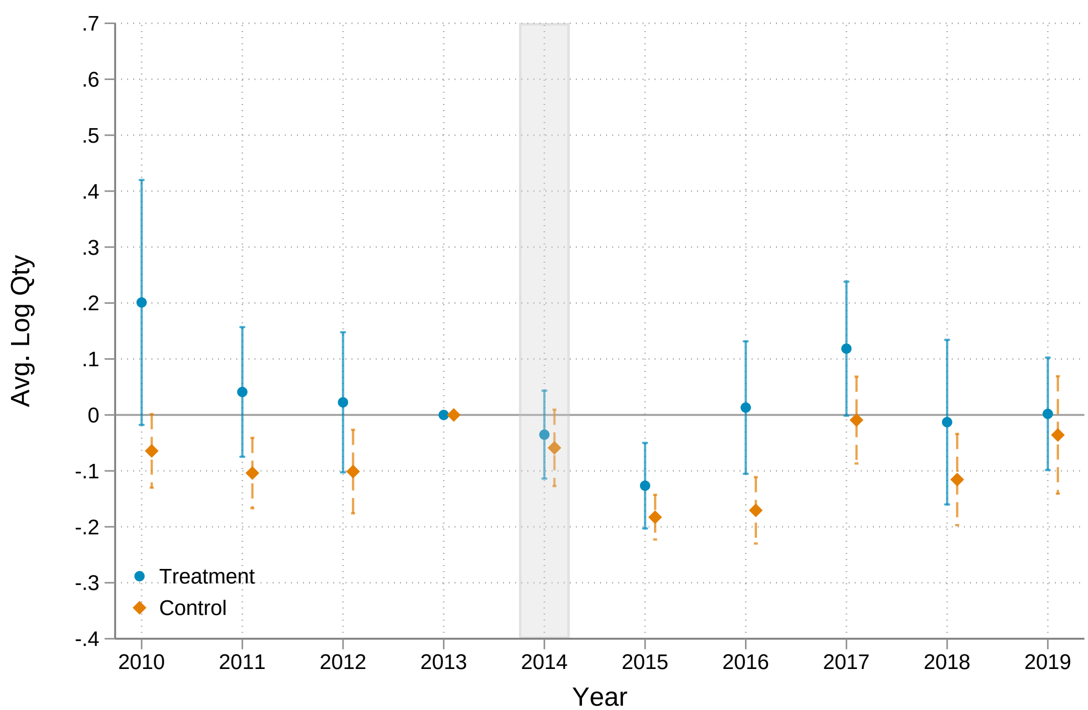
Notes: Naive event study estimates of average log prices around the Thermo Fisher Scientific–Life Technologies merger. Panel (a) plots the pooled difference between treated product markets and the full set of unmatched control markets, and Panel (b) plots the two groups separately, treated (blue) and unmatched controls (orange). Coefficients are normalized to zero at $t = -1$, with the vertical gray band indicating the merger year (2014). The average price in $t = -1$ is $143.30 in treated markets and $84.84 across the full set of control markets.

Figure A2. Pooled Merger-Induced Quantity Effects, Naive Specification

(a) Pooled difference



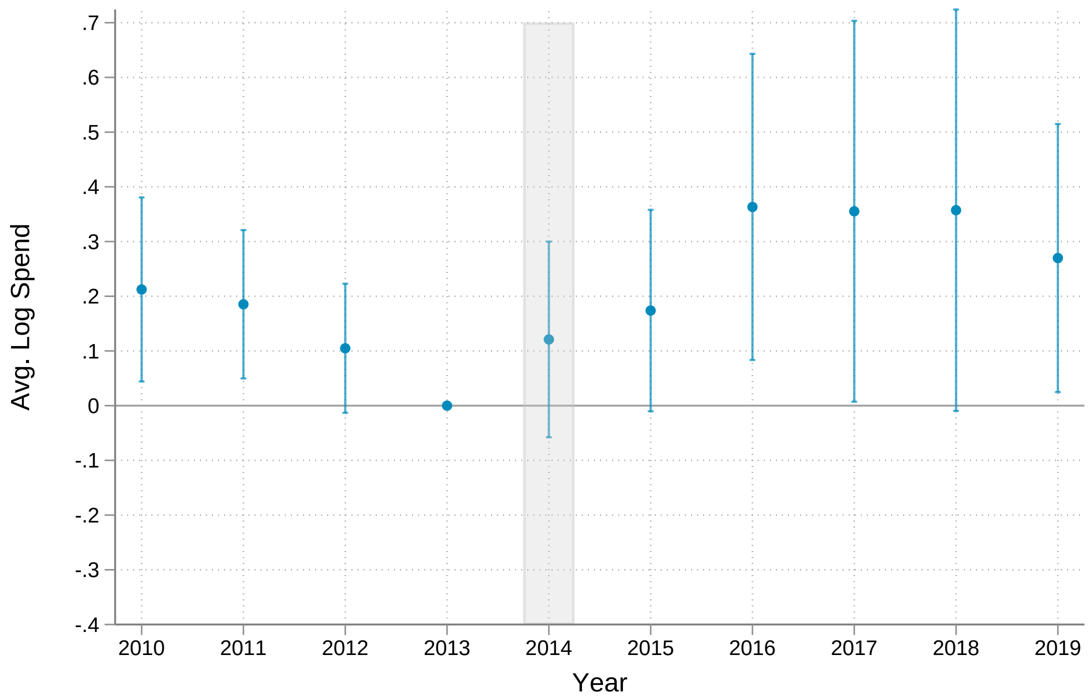
(b) Treated and control series



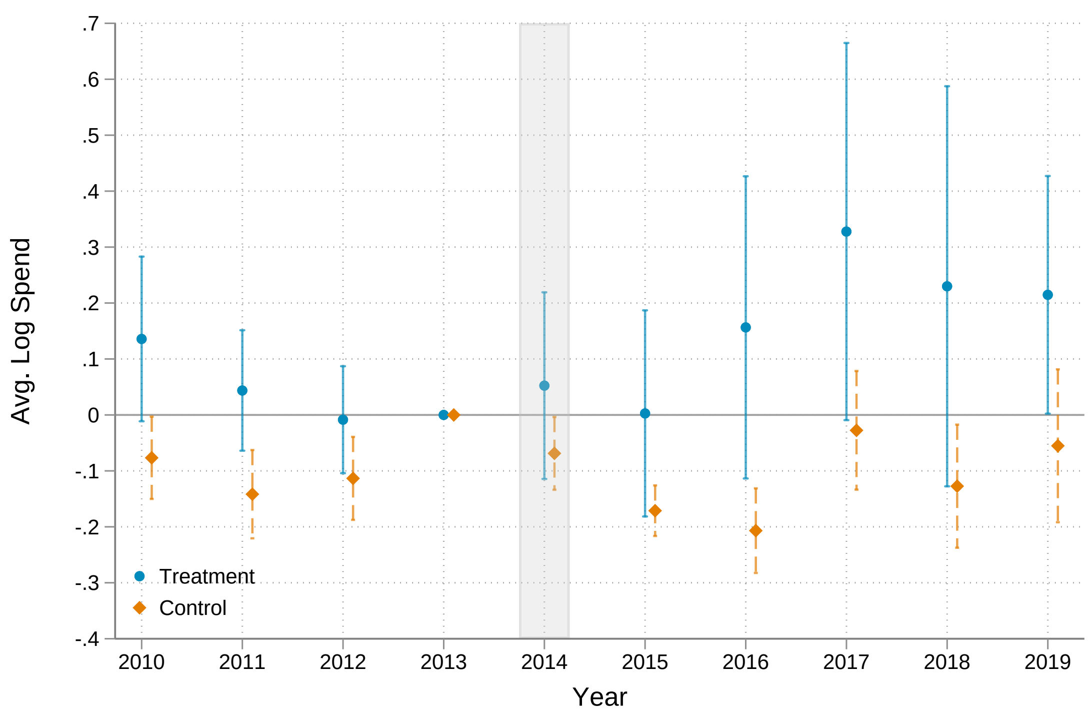
Notes: Naive event study estimates of average log quantity around the merger. Panel (a) plots the pooled difference between treated product markets and the full set of unmatched control markets, and Panel (b) plots the two groups separately, treated (blue) and unmatched controls (orange). Coefficients are normalized to zero at $t = -1$, with the vertical gray band indicating the merger year (2014). The average quantity in $t = -1$ is 17.8 purchases in treated markets and 30.0 across the full set of control markets.

Figure A3. Pooled Merger-Induced Spending Effects, Naive Specification

(a) Pooled difference

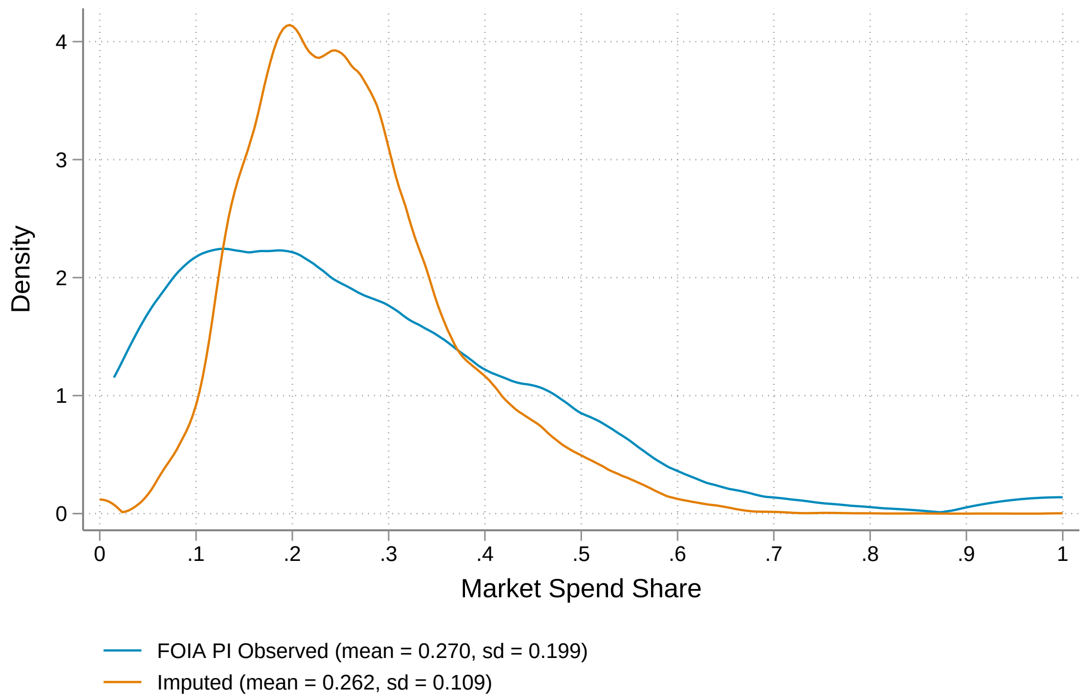


(b) Treated and control series



Notes: Naive event study estimates of average log spending around the merger. Panel (a) plots the pooled difference between treated product markets and the full set of unmatched control markets, and Panel (b) plots the two groups separately, treated (blue) and unmatched controls (orange). Coefficients are normalized to zero at $t = -1$, with the vertical gray band indicating the merger year (2014). Average spending in $t = -1$ is \$2,094 in treated markets and \$1,803 across the full set of control markets.

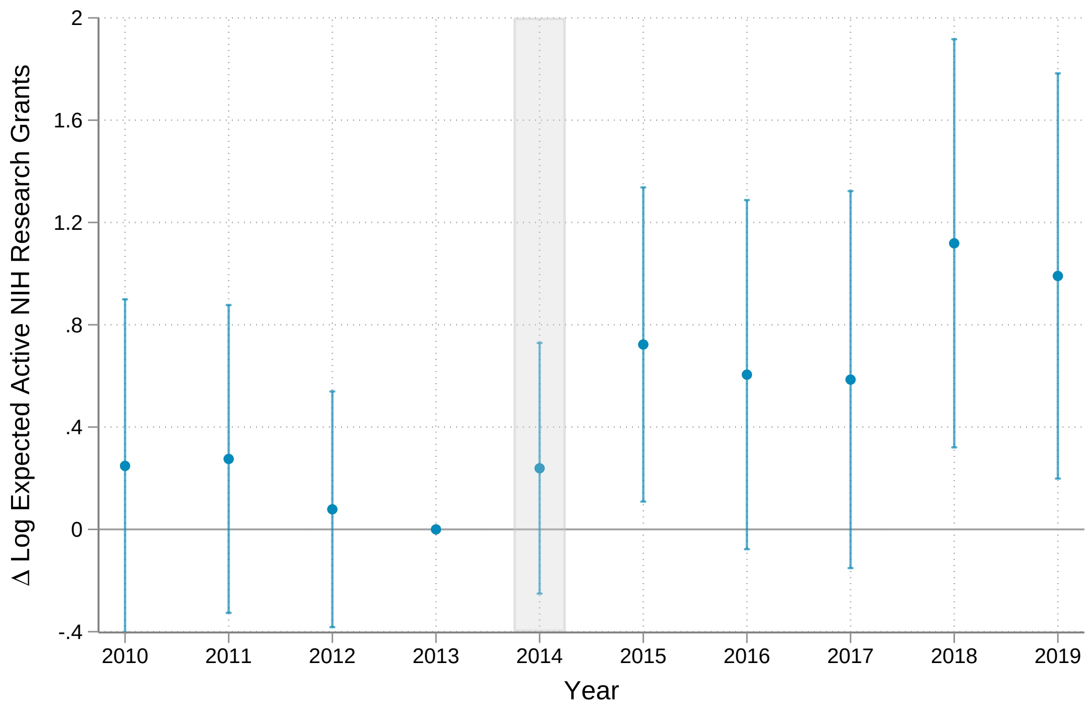
Figure A4. Observed and Imputed Treated-Market Share Distributions



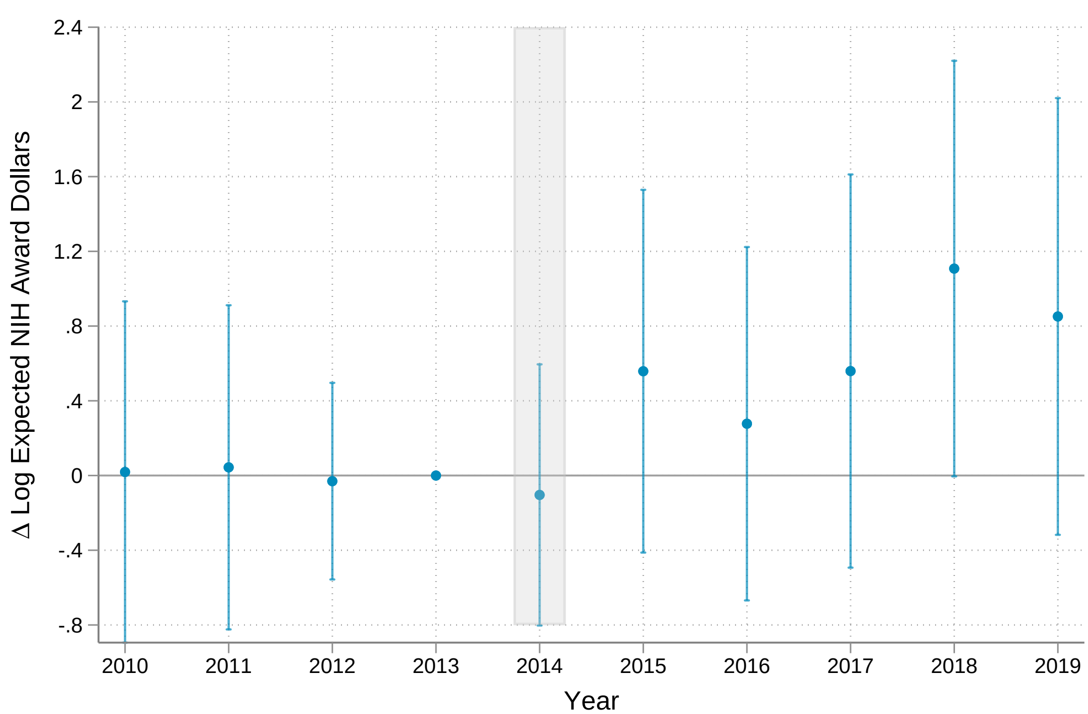
Notes: The figure plots the distribution of the treated-market spending share $S_i = \sum_m s_{im}$, observed directly for the 208 FOIA PIs (blue) and imputed for the 13,433 non-FOIA PIs in the panel (orange). S_i sums a PI's pre-merger spending shares across the high-confidence treated markets and enters the event study of equation (4) as a control.

Figure A5. Effect of the Input Cost Shock on NIH Funding

(a) Active NIH research grants



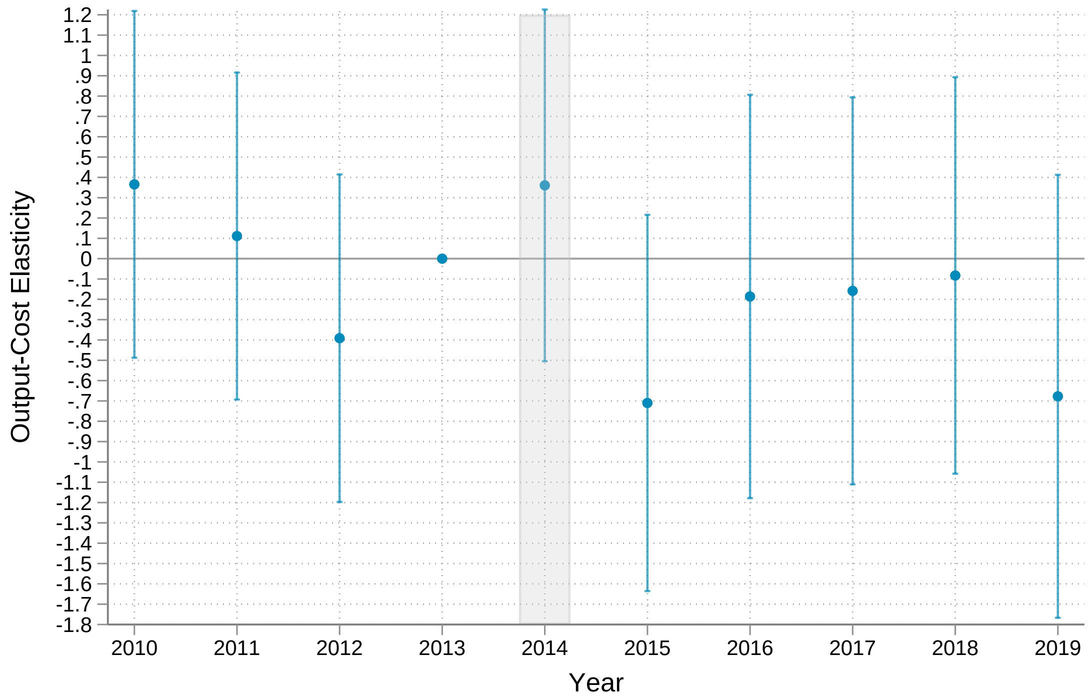
(b) NIH award dollars



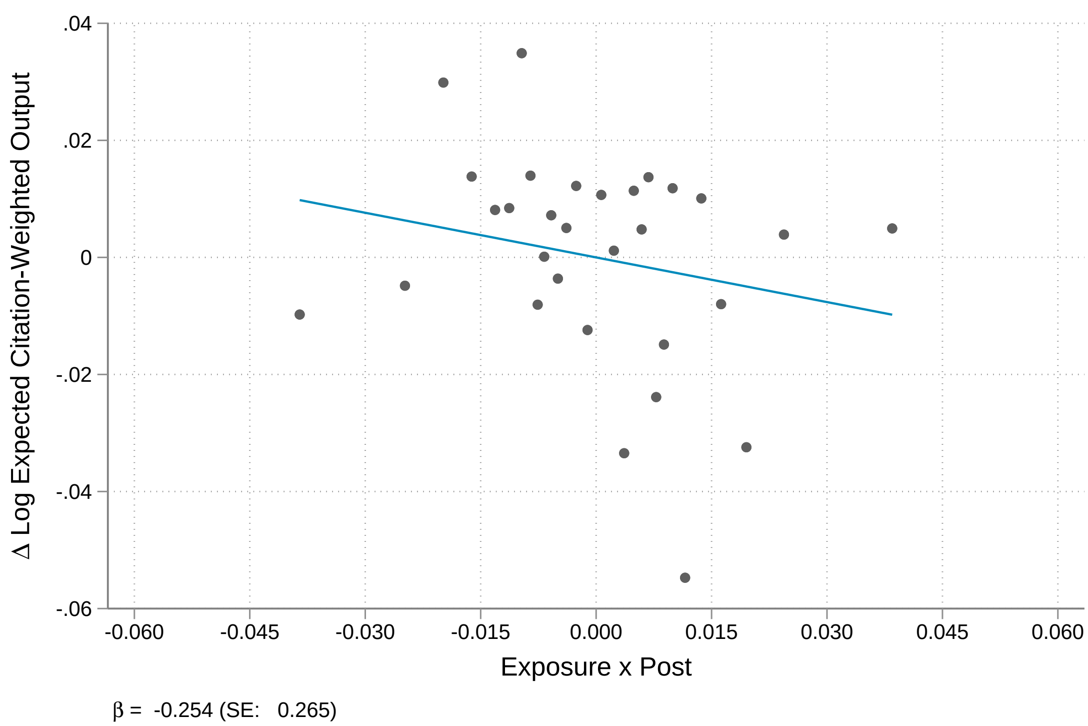
Notes: Each panel plots event study estimates from equation (4) with a measure of NIH funding as the outcome, normalized to zero at $t = -1$, with the vertical gray band indicating the merger year (2014), for the 7,530 panel PIs matched to NIH records at 183 universities. Panel (a) is the number of active NIH research grants, with a pre-merger mean of 1.27 per PI, and Panel (b) is annual NIH award dollars, with a pre-merger mean of $496,000.

Figure A6. Effect of the Input Cost Shock on Citation-Weighted Output

(a) Event study

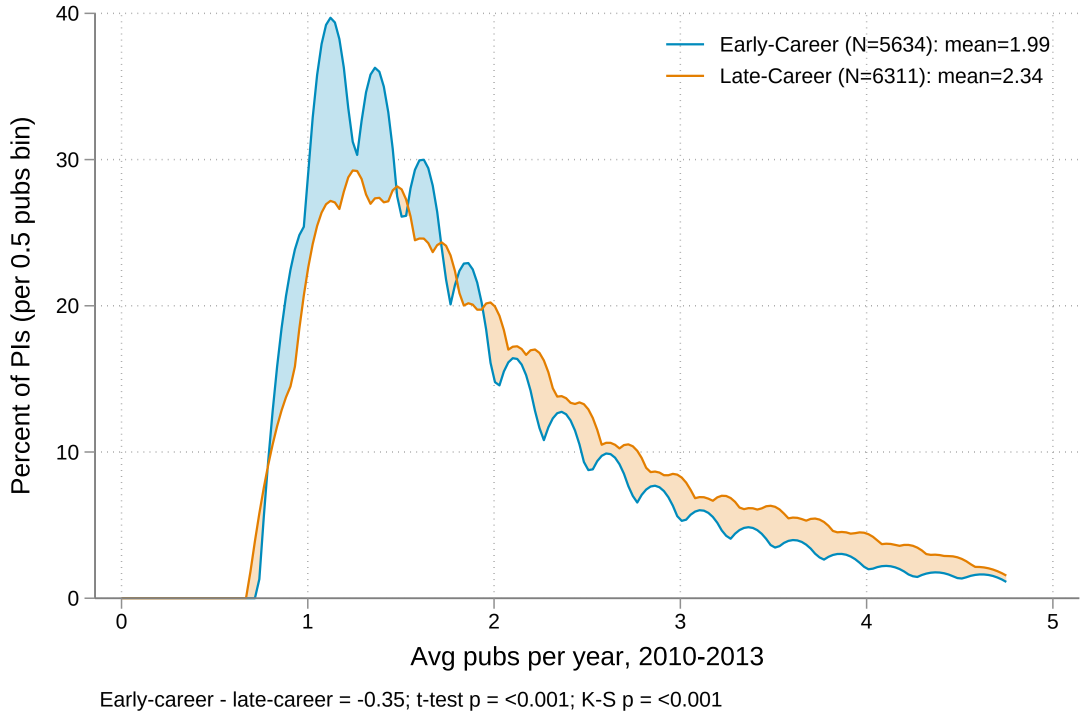


(b) Difference-in-differences



Notes: Panel (a) plots event study estimates from equation (4) with citation-weighted output as the outcome, normalized to zero at $t = -1$. Panel (b) is a binned scatter of citation-weighted output against $\text{Post}_t \times \text{Exposure}_i$ after partialling out PI and year fixed effects and the treated-market share control, with fitted slope $\beta = -0.254$ (SE 0.265). The measure weights each paper by its average annual citations and each author's effective contribution and rescales the weighted counts to sum to the total number of papers. The sample is the baseline panel of 13,515 PIs, with a pre-merger mean of 4.26 per PI-year.

Figure A7. Pre-Merger Publication Rates by Career Stage



Notes: The figure plots kernel densities of average annual publications over 2010–2013 for early-career and late-career PIs with at least one publication in that period. Early-career PIs average 1.99 publications per year and late-career PIs 2.34, a difference statistically significant at the 1% level.

B High-Confidence Markets

This appendix lists the 152 product markets that our NLP classifier assigns with high confidence and that enter the estimation of the price effects, after the 12 control markets subject to contemporaneous shocks described in Section 3.2 are removed. This list consists of 46 treated markets and 106 control markets. A market is classified as high-confidence if it has at least 25 training examples and precision and recall of at least 0.8. Treated markets are those the European Commission identified as having competitive overlap in its clearance decision (Case COMP/M.6944), and control markets fall outside the merging firms' direct overlap. Within each group, the markets are organized into broad categories for readability.

Treated Markets

- **Cell culture media, sera, and supplements.** DMEM, DMEM/F-12, RPMI, U.S.-origin fetal bovine serum, donkey serum, goat serum, horse serum, Dulbecco's phosphate-buffered saline (DPBS), Hanks' balanced salt solution (HBSS), cell-culture dissociation reagents, tryptone, and yeast extract.
- **Nucleic-acid amplification and modifying enzymes.** Taq DNA polymerase, high-fidelity hot start DNA polymerase, hot-start PCR systems, reverse transcriptase, first-strand cDNA synthesis systems, DNase I, RNase inhibitors, Klenow fragment, T4 polynucleotide kinase, and site-directed mutagenesis systems.
- **Nucleic-acid purification kits.** Column-based plasmid miniprep, midiprep, and maxiprep kits, gel DNA extraction kits, microbial DNA purification kits, and RNA stabilization reagent.
- **Gel electrophoresis and protein blotting.** Acrylamide-bis solution, pre-cast Bis-Tris gels, pre-cast TBE gels, pre-cast Tris-glycine gels, pre-cast Tris-tricine gels, MES-SDS running buffer, Laemmli sample buffer, LDS sample buffer, nitrocellulose blotting membranes, PVDF blotting membranes, pre-stained protein molecular-weight ladders, unstained DNA ladders, nucleic-acid gel stains, protein gel stains, and chemiluminescent substrates.
- **General buffers and blocking reagents.** Phosphate-buffered saline (PBS), Tris-EDTA (TE) buffer, and bovine serum albumin.

Control Markets

- **Bulk chemicals, acids, and solvents.** Acetone, chloroform, ethyl acetate, isopropanol, methanol, tetrahydrofuran (THF), trifluoroacetic acid (TFA), xylenes, phenols, glycerol, glycine, nitric acid, sulfuric acid, boric acid, hydrogen peroxide, bleach, ammonium chloride, ammonium persulfate, lithium chloride, magnesium chloride, potassium carbonate, potassium phosphate, sodium acetate, sodium azide, sodium bicarbonate, sodium chloride, sodium citrate, sodium hydroxide, sodium sulfate, and zinc chloride.
- **Buffering agents.** HEPES buffer, MES buffer, and MOPS buffers.
- **Antibiotics and selection reagents.** Ampicillin, carbenicillin, chloramphenicol, hygromycin, penicillin-streptomycin, puromycin, zeocin, and IPTG.

- **Detergents, lysis, and reducing reagents.** Digitonin, SDS (sodium dodecyl sulfate), Triton X-100, Tween detergents, DTT, TCEP, and protease inhibitor cocktails.
- **Stains, substrates, and detection kits.** Apoptosis detection kits, bioluminescence assay kits, colorimetric substrates (DAB, MTT, and TMB), histology and cell stains, phalloidin conjugates, viability stains, western blot stripping buffers, and PAP pens.
- **Separation media and other molecular-biology reagents.** Agarose, dNTPs, GST-tag affinity resins, size-exclusion resins, spin desalting columns, and dialysis cassettes.
- **Cell-culture and microbiology reagents.** Carbon dioxide, laminin, polylysines, thrombin, dehydrated bacterial broth, and inoculating loops and needles.
- **Labware, plasticware, and glassware.** Cell culture dishes, flasks, plates, and tubes, coverslips, microscope slides, cell scrapers, cell strainers, cryovials, cuvettes, beakers, boiling flasks, dewar flasks, disposable pipettes, serological pipettes, pipette tips, extraction thimbles, film and foil wrapping, filter papers, funnels, graduated cylinders, lab coats, laboratory spatulas, latex gloves, media bottles, microcentrifuge tubes, PCR tube strips, petri dishes, spray bottles, square glass bottles, stir bars, storage jars, vials, wash bottles, weigh boats, and weighing papers.

C Survey of Life Sciences PIs

We run a short survey of active U.S. life sciences PIs about how they manage supply costs. We recruit respondents by email in two stages. We first ran a pilot of 54 PIs across ten universities, drawn in equal numbers from assistant, associate, and full professors to gauge response rates. The main wave begins with large R1 and R2 universities. At each university we select up to three life sciences departments, such as molecular biology, immunology, and biological sciences, collect the faculty listed on each department's website with the assistance of a large language model (Claude Opus 5), and email invitations to a random half of the listed faculty. We began sending main-wave invitations on August 13, 2026. Across the two stages, we have emailed 255 PIs at 14 public and private institutions as of August 31, 2026. The 19 responses received imply a response rate of about 7 percent.

The survey is administered as an online questionnaire through Google Forms and was deemed exempt by the Harvard University IRB (IRB26-0945). Data collection is ongoing, so we treat these results as preliminary and suggestive. The full survey instrument is reproduced on the following pages.

* Indicates required question

About your lab

1. What year did you start running your own lab? *

2. What best describes your field? *

Check all that apply.

- [ ] Life sciences, mainly experimental (bench work)
- [ ] Life sciences, mainly computational
- [ ] Chemistry
- [ ] Engineering, physics, or materials science
- [ ] Prefer not to say
- [ ] Other:

3. Which best describes your the institution your lab is based in? *

Mark only one oval.

- Public University
- Private University
- Academic medical center or hospital
- Nonprofit or independent research institute
- Prefer not to say
- Other:

4. How many people work in your lab, including you? Type NA if you'd rather not answer. *

____

How your lab buys supplies

5. Who places most supply orders for your lab? *

Mark only one oval.

- ( ) I do
- ( ) A lab manager or technician
- ( ) Students and postdocs order what they need
- ( ) A purchasing office
- ( ) Prefer not to say
- ( ) Other: ____

6. Which of the following does your lab currently do? Select all that apply. *

Check all that apply.

- [ ] Compare prices across suppliers before ordering
- [ ] Request quotes or negotiate prices with vendors
- [ ] Order through your institution's purchasing system to get negotiated prices
- [ ] Maintain a standing list of preferred products or vendors
- [ ] Track supply spending against the budget regularly
- [ ] Set rules about what lab members can order without approval
- [ ] None of these
- [ ] Prefer not to say
- [ ] Other: _______________

7. Now think back to your first two years running your own lab. Which of these did you do then? Select all that apply. *

Check all that apply.

- [ ] Compare prices across suppliers before ordering
- [ ] Request quotes or negotiate prices with vendors
- [ ] Order through your institution's purchasing system to get negotiated prices
- [ ] Maintain a standing list of preferred products or vendors
- [ ] Track supply spending against the budget regularly
- [ ] Set rules about what lab members can order without approval
- [ ] None of these
- [ ] I do not remember
- [ ] Prefer not to say
- [ ] Other: ____________________

8. How did you learn to manage supply purchasing for a lab? Select all that apply. *

Check all that apply.

- [ ] Trial and error once I had my own lab
- [ ] From my PhD or postdoc advisor
- [ ] From a lab manager or technician
- [ ] From colleagues or other PIs
- [ ] Institutional training or onboarding
- [ ] Someone else handles it
- [ ] I have not really learned it
- [ ] Prefer not to say
- [ ] Other: ____________________

When costs go up

9. Think of the most recent time the price of something your lab buys went up noticeably. What did your lab do? Select all that apply. *

Check all that apply.

- [ ] Switched to a cheaper supplier or brand
- [ ] Bought in larger quantities, or coordinated orders with other labs
- [ ] Made reagents or solutions in-house instead of buying pre-made kits
- [ ] Used less material per experiment (smaller volumes, fewer replicates)
- [ ] Ran fewer experiments, or dropped conditions
- [ ] Delayed or reduced hiring (postdoc, technician, student support)
- [ ] Delayed or reduced equipment purchases
- [ ] Shifted costs onto a different grant or fund
- [ ] Applied for additional funding
- [ ] Nothing, we absorbed it
- [ ] This has not happened, or I have not noticed it happening
- [ ] Prefer not to say
- [ ] Other: ______________________

10. By roughly how much would the price of something you buy regularly have to rise before you changed something about how the lab works? *

Mark only one oval.

- ◯ Less than 5%
- ◯ 5 to 10%
- ◯ 10 to 25%
- ◯ 25 to 50%
- ◯ More than 50%
- ◯ I would not change anything
- ◯ Prefer not to say

11. Since you started running your lab, has a change in supply costs ever affected how you work? If so, describe what happened and what you did. Anything else you would like to add about supply costs or purchasing is welcome here. *

12. Roughly what does your lab spend per year on consumable supplies and reagents? Please exclude salaries, instruments and equipment, and core facility fees (sequencing, mass spec, imaging). *

Mark only one oval.

- Under $10,000
- $10,000 to $49,999
- $50,000 to $149,999
- $150,000 to $499,999
- $500,000 or more
- Not sure
- Prefer not to stay
- Other:

13. Roughly what percentage of your lab's total annual budget does that represent? Include salaries and all other direct costs, but exclude indirect or overhead costs. Type NA if you'd rather not answer. *

14. Thermo Fisher acquired Life Technologies (Invitrogen, Applied Biosystems, Gibco) * in early 2014. Did you notice any price change in those brands at the time?

Mark only one oval.

- Yes, noticeable increase
- Yes, slight increase
- No change I noticed
- I was not running a lab in 2014
- I did not buy those brands
- Do not remember
- Prefer not to say
- Other:

15. We are also speaking with PIs for about 30 minutes about how labs manage supply costs. If you would be willing to have that conversation, please leave your email address. This is optional, and your address will be stored separately from your responses and used only to contact you about an interview.